Continuous coupling trajectories — run report

RUN

220235_cross-sweep_phi2-rerun
Status: completed | Started: 2026-09-29T21:02:35.273554+00:00

DIAGNOSTIC SUMMARY

Target actuator(s): phi2; sine: 0.1164 Hz; measured PAX rate: 6.87 Hz; saved RP reference rate: 381 Hz; estimated PAX samples/cycle: 58.98.
Recorded cycles: 3; bias values: phi1 target 0, phi2 target 11. Waveform tracking: correct within descriptive threshold. Rising/falling branches (>5° descriptive threshold): strong separation present. Cycle repeatability (5° descriptive threshold): nonrepeatable above threshold. Bias-dependent plane changes (>5° descriptive threshold): obvious changes present. Low-DoP intervals: True. These thresholds are flags, not mechanism assignments.

PURPOSE

Continuously sine-drive phi1, phi2, or both while stepping and settling the orthogonal bias; record PAX plus measured OUT1/OUT2 references.

REQUIRED OPTICAL SETUP

PAX at final output; both paths open unless prompted. Wire OUT1 to IN1 and OUT2 to IN2 as voltage references. 'both' runs the complete diagnostic for both actuators automatically.

RECORDED BENCH CONTEXT (RECIPE)

In front of NPBS 1, C port
Red Pitaya (0-1 V command) -> Thorlabs MDT690 -> piezo

RUN COMMENT

same as phi1

OPERATOR SETUP NOTES

V_pi expectation only; verify channel and measured driver gain before calibration.

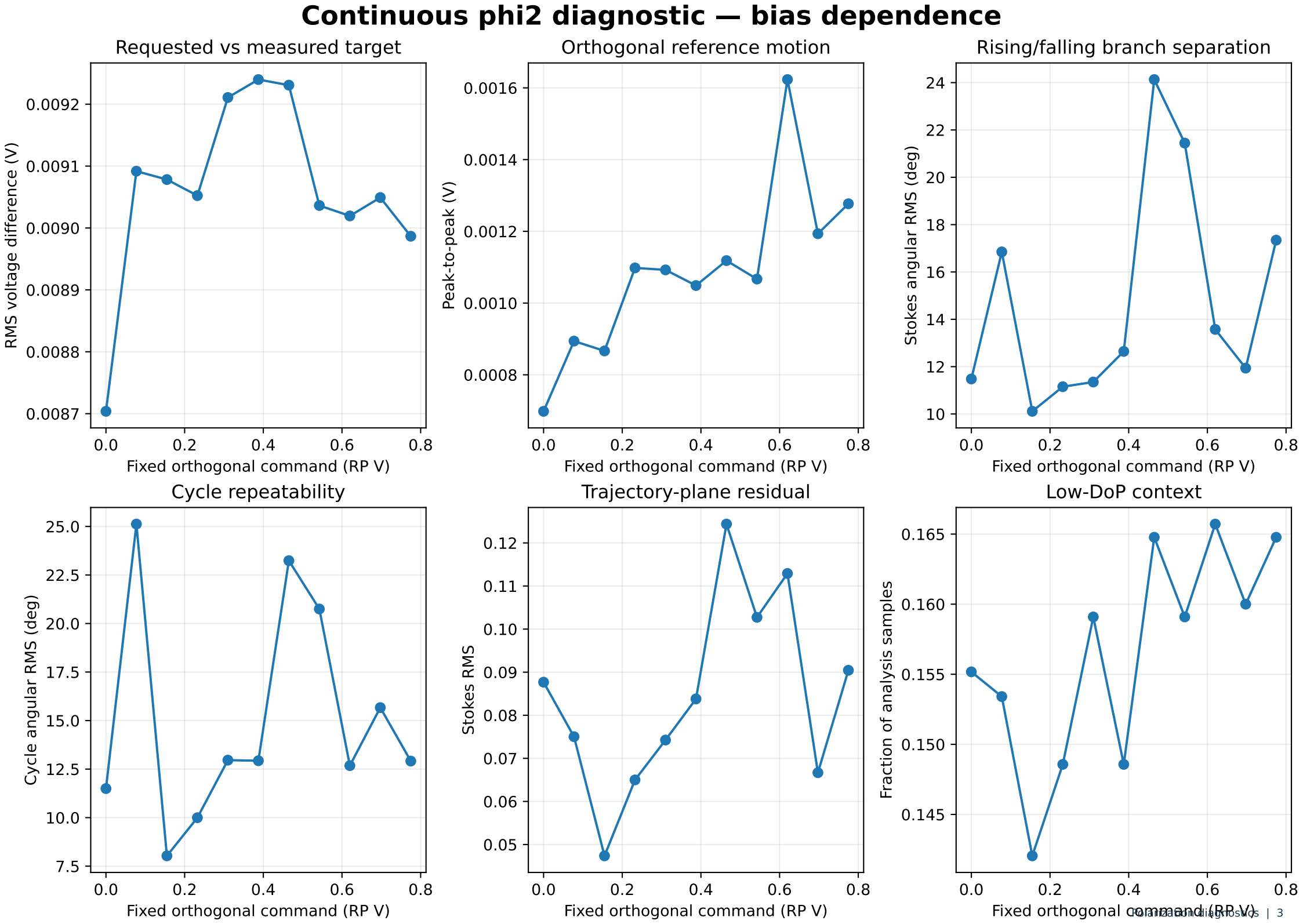
DATA

Continuous coupling trajectories — run report

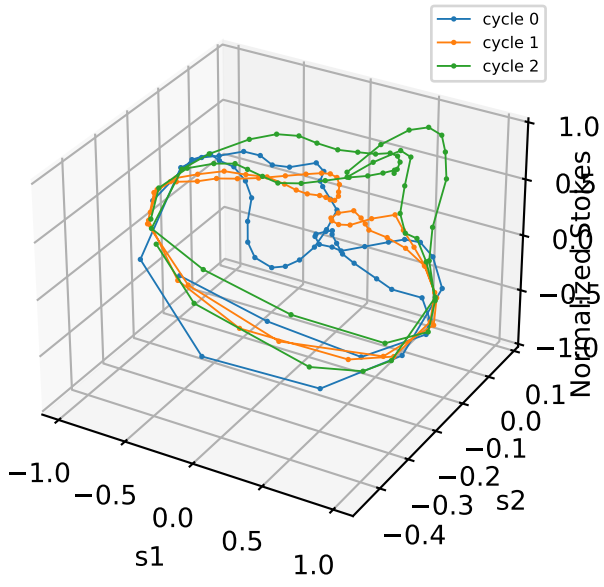
2,669 recorded samples; acquisition: 2,669
Source: data.csv; full acquisition settings: recipe.json; status: run.json

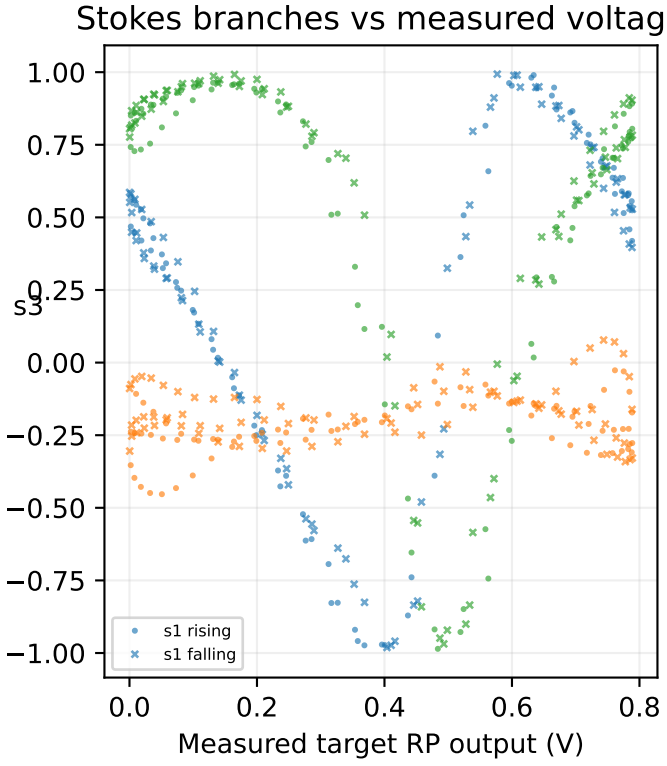
RAW DOP

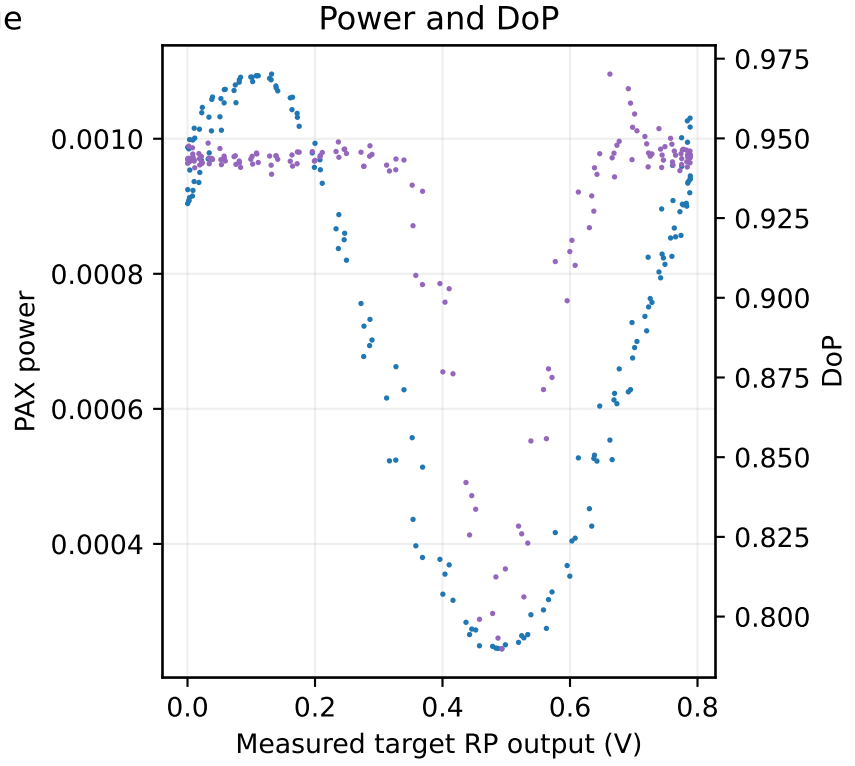
Mean 0.9271; range 0.1975 to 1.1460. 2 samples outside 0–1. Telemetry is not a new calibration.

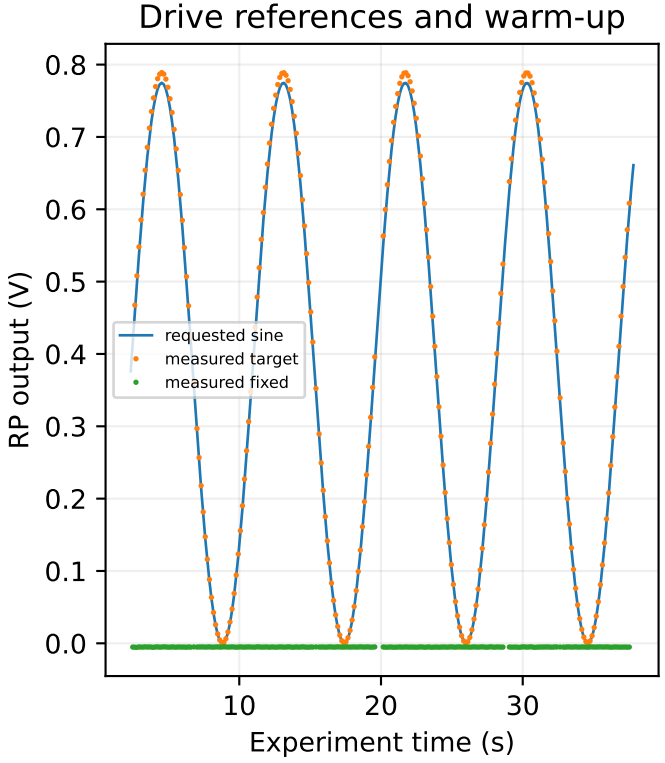

Continuous phi2 diagnostic — bias dependence
Requested vs measured target
RMS voltage difference (V)
Fixed orthogonal command (RP V)
Orthogonal reference motion
Peak-to-peak (V)
Fixed orthogonal command (RP V)
Rising/falling branch separation
Stokes angular RMS (deg)
Fixed orthogonal command (RP V)
Cycle repeatability
Cycle angular RMS (deg)
Fixed orthogonal command (RP V)
Trajectory-plane residual
Stokes RMS
Fixed orthogonal command (RP V)
Low-DoP context
Fraction of analysis samples
Fixed orthogonal command (RP V)

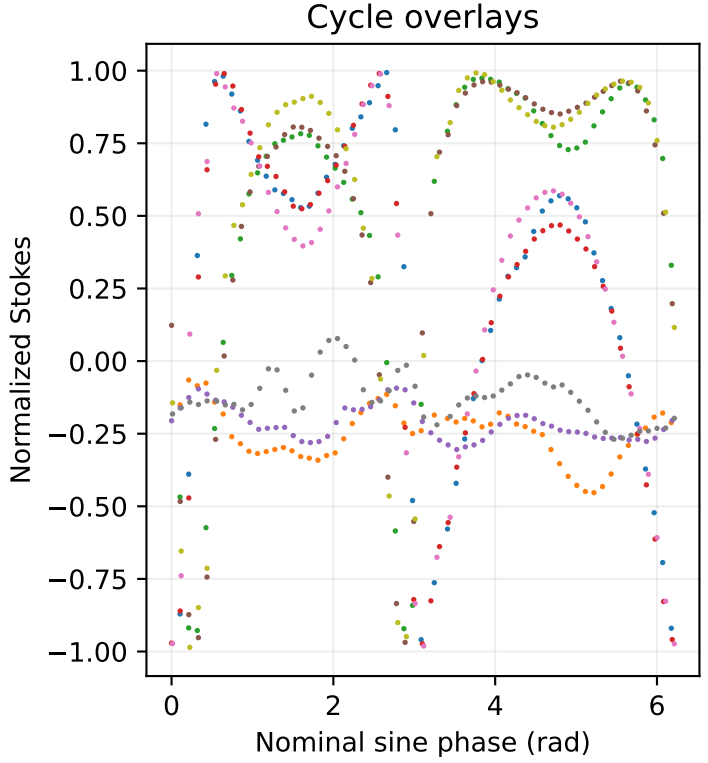
phi2 sine; fixed phi1 = 0 RP V (segment 0)





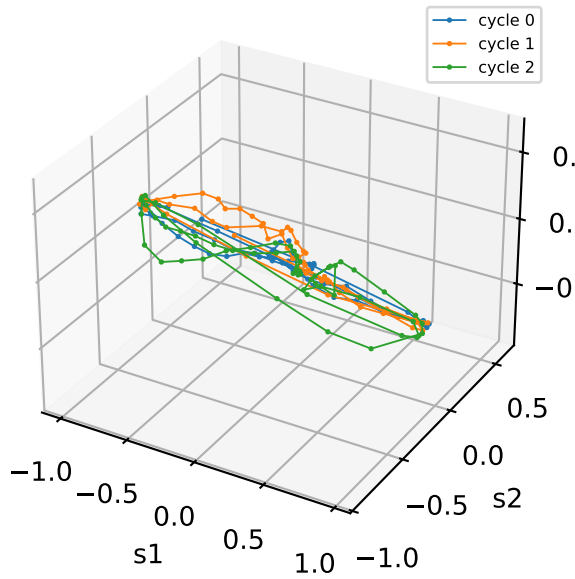


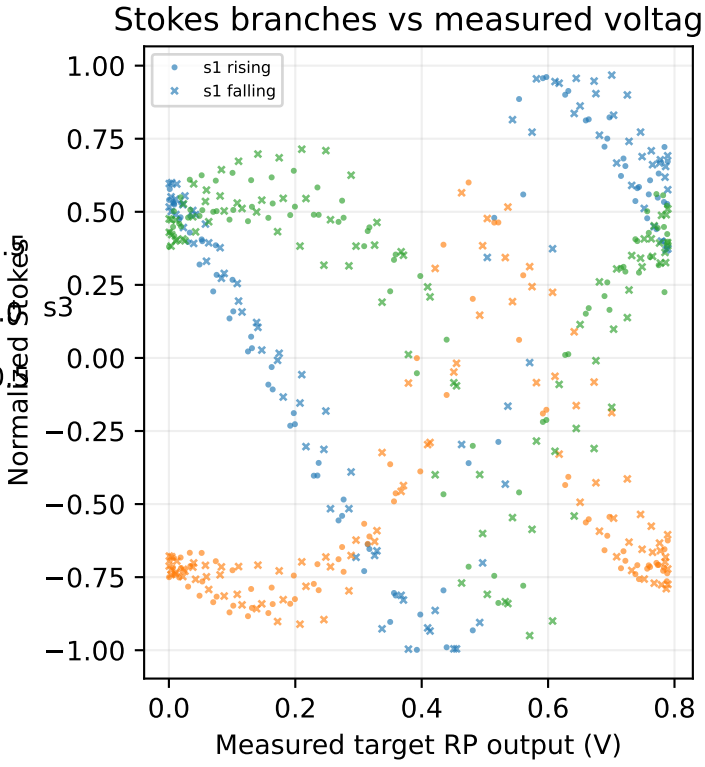


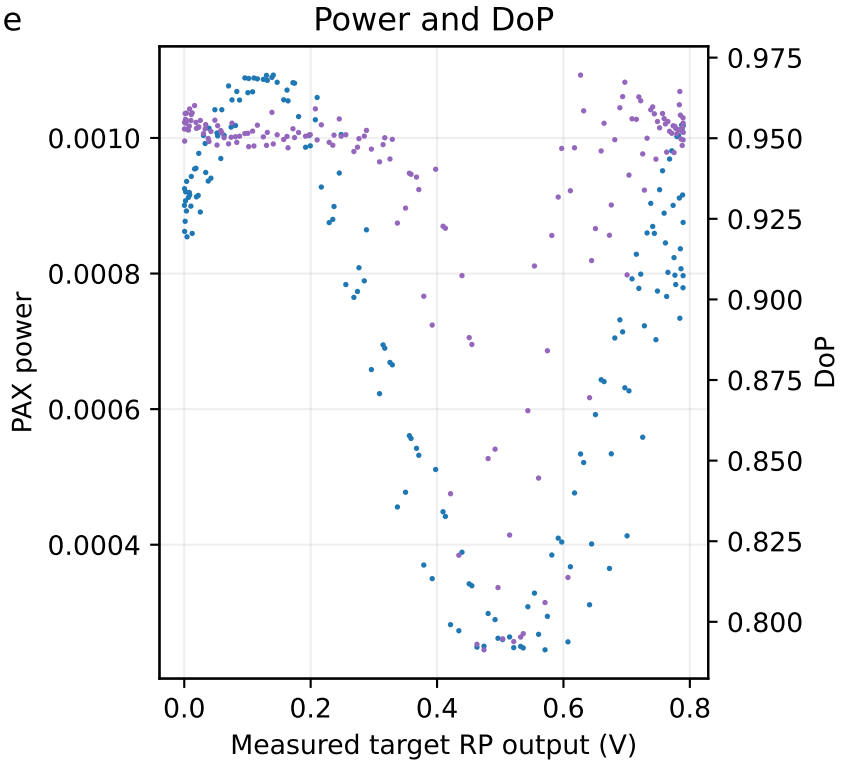


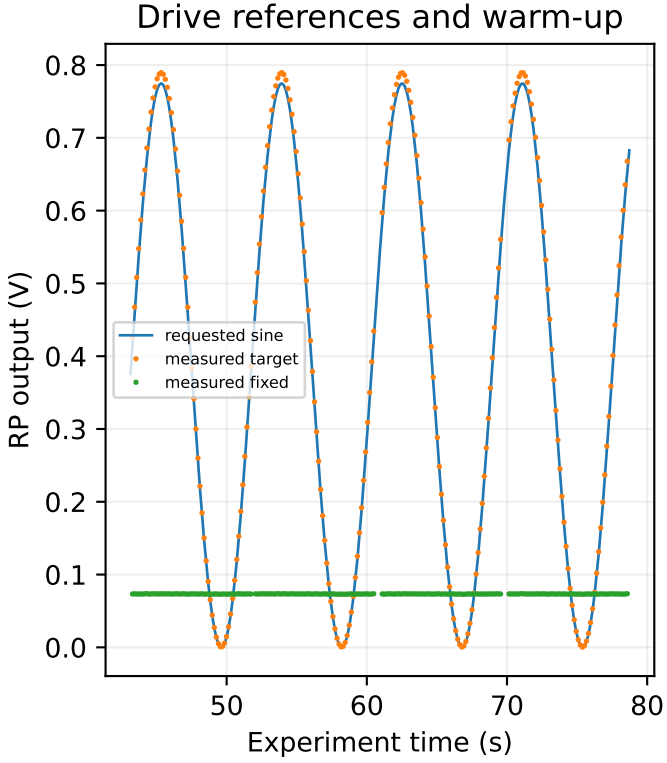
Ideal reference: phi2 changes polar coordinate; fixed phi1 sets azimuth orientation.
Plane normal: [-0.026, 0.998, 0.060]
Plane residual RMS: 0.08768
Rising/falling angular RMS: 11.5°
Rising/falling power RMS: 5.5e-05 PAX units
Cycle angular RMS: 11.5°
Cycle power RMS: 4.3e-05 PAX units
Cycle DoP RMS: 0.0111
Tracking RMS: 0.008704 V
Reference coverage: 97.1%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.03, 1.00, 0.09], residual 0.0761
Cycle 1 plane: [-0.03, 1.00, 0.09], residual 0.0243
Cycle 2 plane: [-0.07, 1.00, 0.01], residual 0.0732

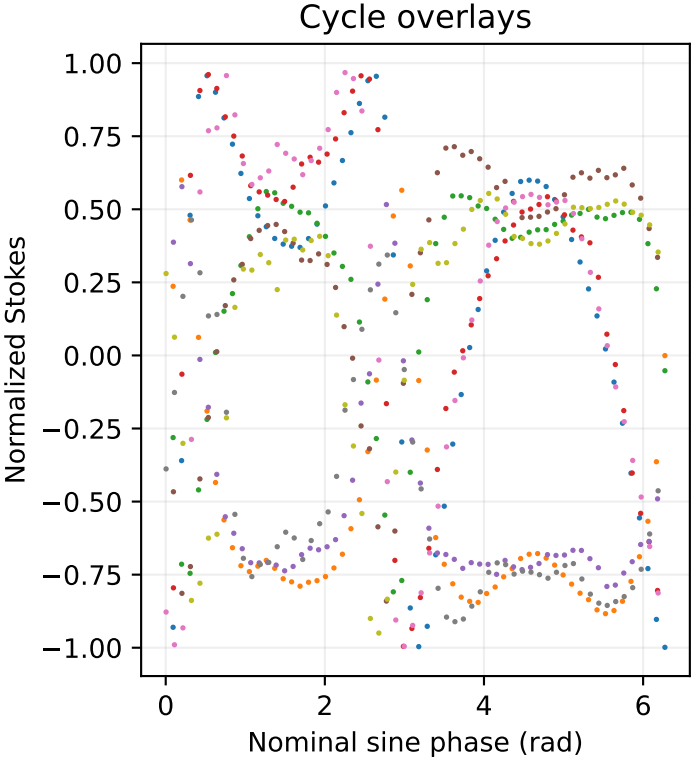
phi2 sine; fixed phi1 = 0.07739 RP V (segment 1)





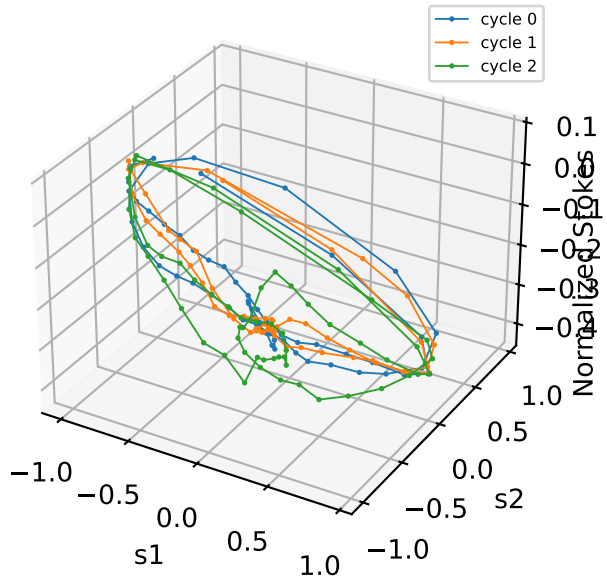


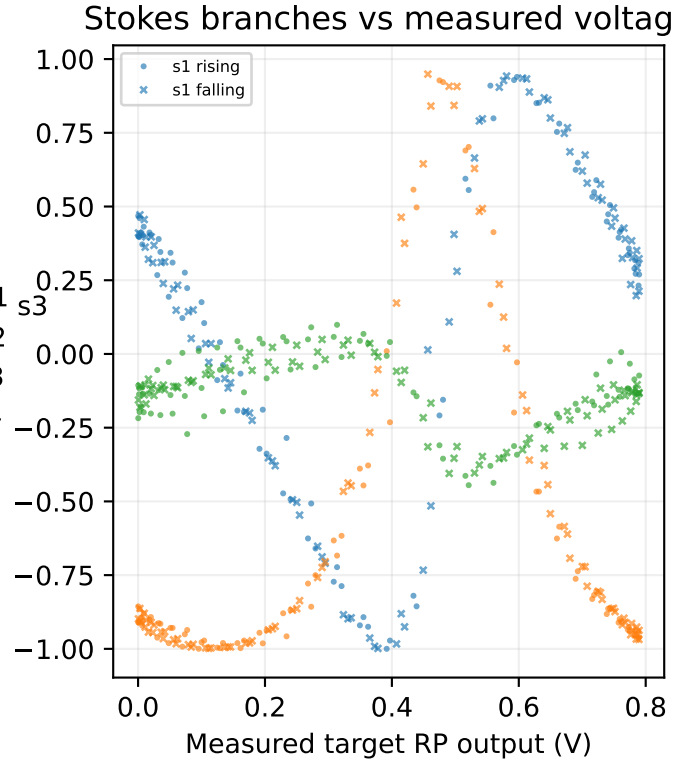


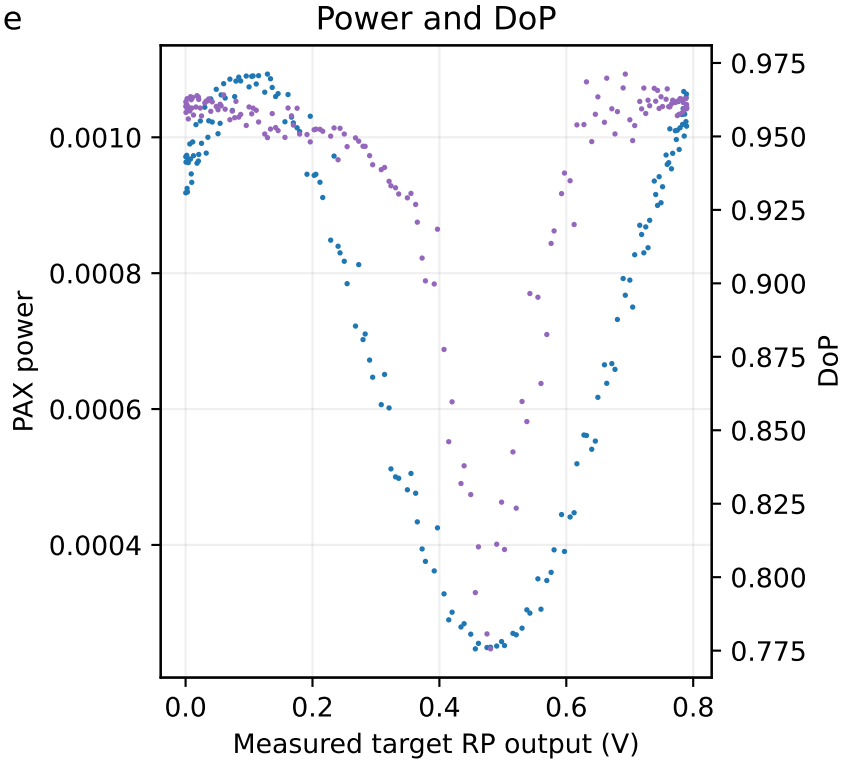


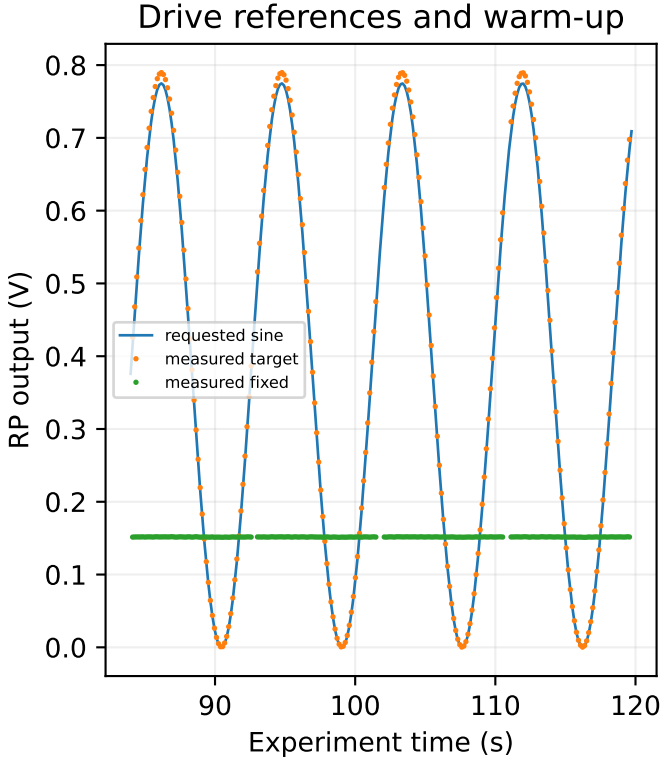
Ideal reference: phi2 changes polar coordinate; fixed phi1 sets azimuth orientation.
Plane normal: [0.090, 0.734, 0.673]
Plane residual RMS: 0.07503
Rising/falling angular RMS: 16.8°
Rising/falling power RMS: 7.55e-05 PAX units
Cycle angular RMS: 25.1°
Cycle power RMS: 0.00012 PAX units
Cycle DoP RMS: 0.029
Tracking RMS: 0.009092 V
Reference coverage: 96.0%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.09, 0.69, 0.72], residual 0.0446
Cycle 1 plane: [0.12, 0.76, 0.64], residual 0.0545
Cycle 2 plane: [0.07, 0.75, 0.66], residual 0.083

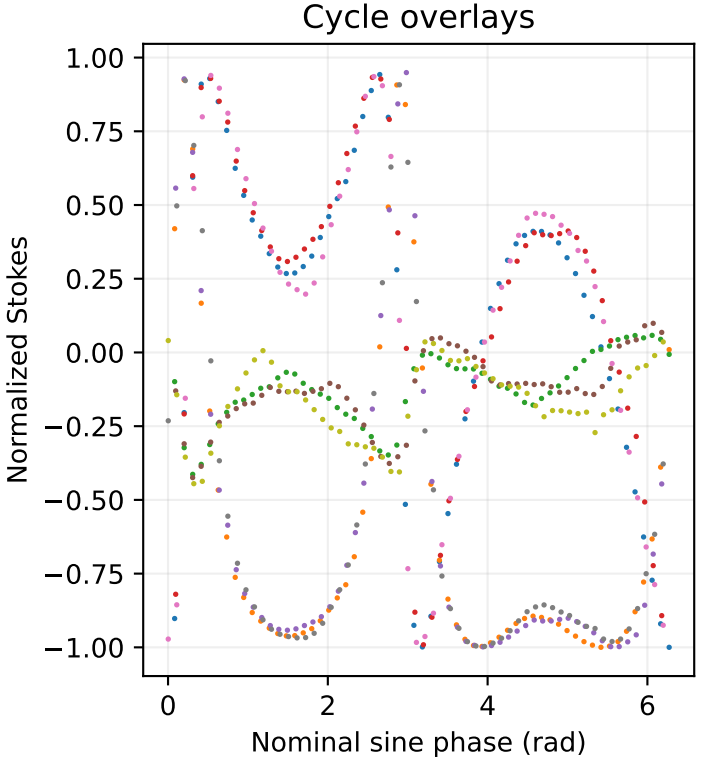
phi2 sine; fixed phi1 = 0.1549 RP V (segment 2)





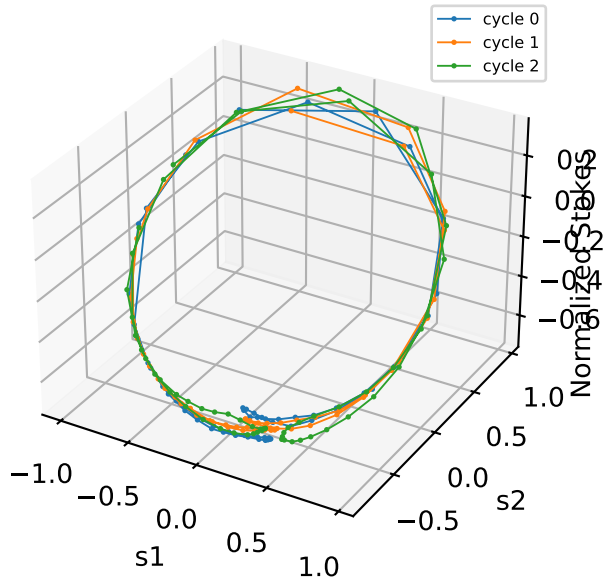


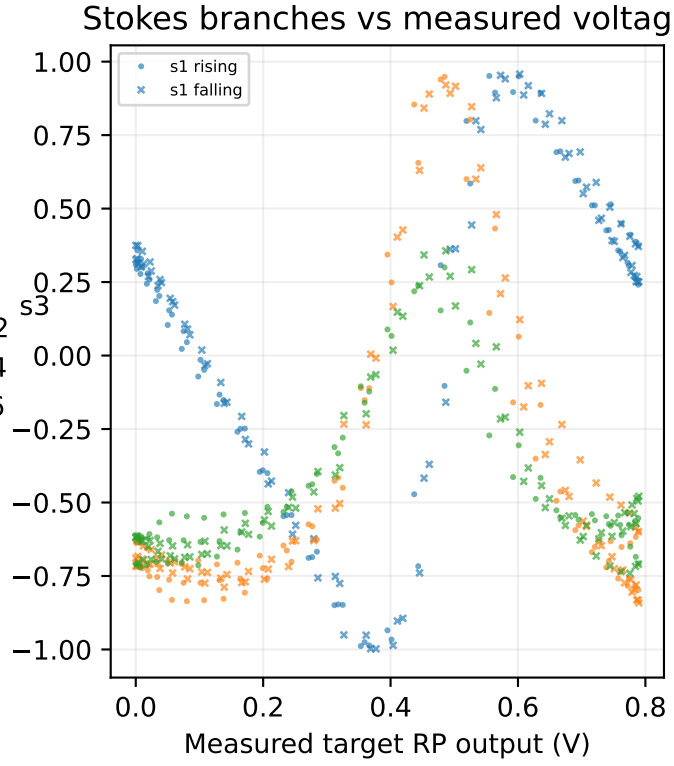


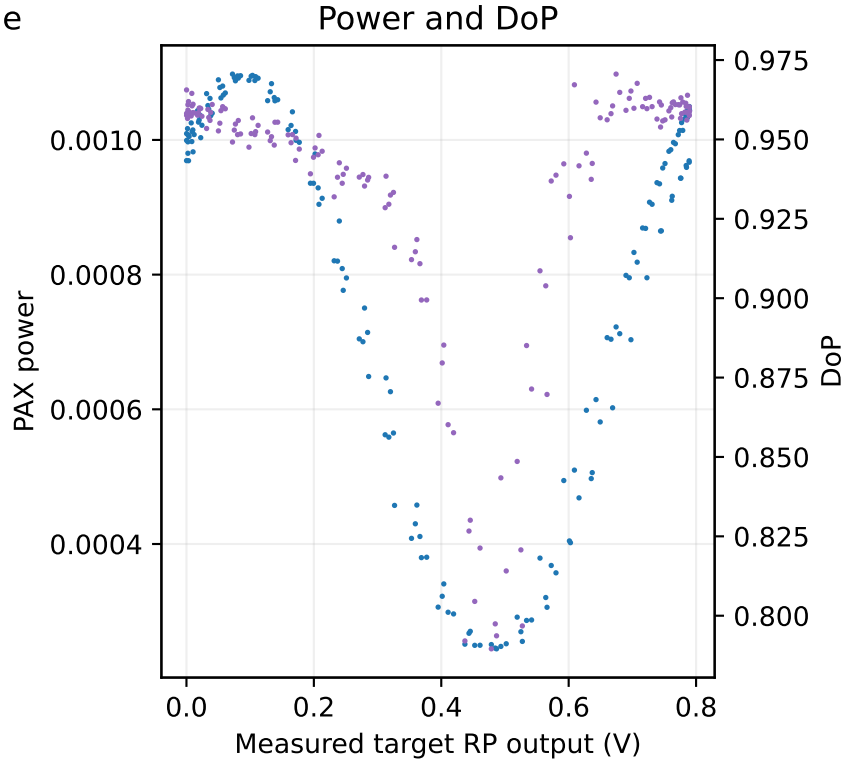


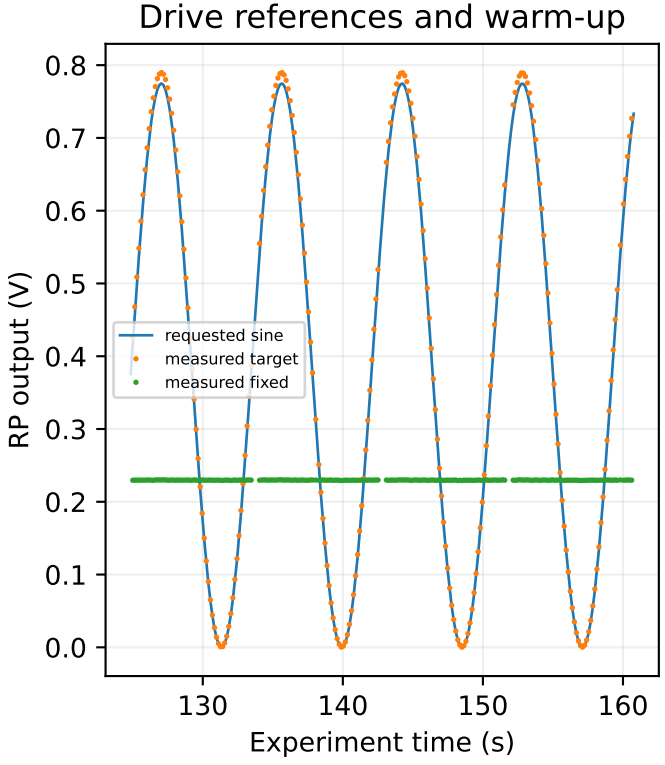
Ideal reference: phi2 changes polar coordinate; fixed phi1 sets azimuth orientation.
Plane normal: [0.166, 0.136, 0.977]
Plane residual RMS: 0.04737
Rising/falling angular RMS: 10.1°
Rising/falling power RMS: 3.96e-05 PAX units
Cycle angular RMS: 8.03°
Cycle power RMS: 3.91e-05 PAX units
Cycle DoP RMS: 0.00952
Tracking RMS: 0.009078 V
Reference coverage: 95.5%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.17, 0.13, 0.98], residual 0.032
Cycle 1 plane: [0.18, 0.14, 0.97], residual 0.0262
Cycle 2 plane: [0.16, 0.13, 0.98], residual 0.0628

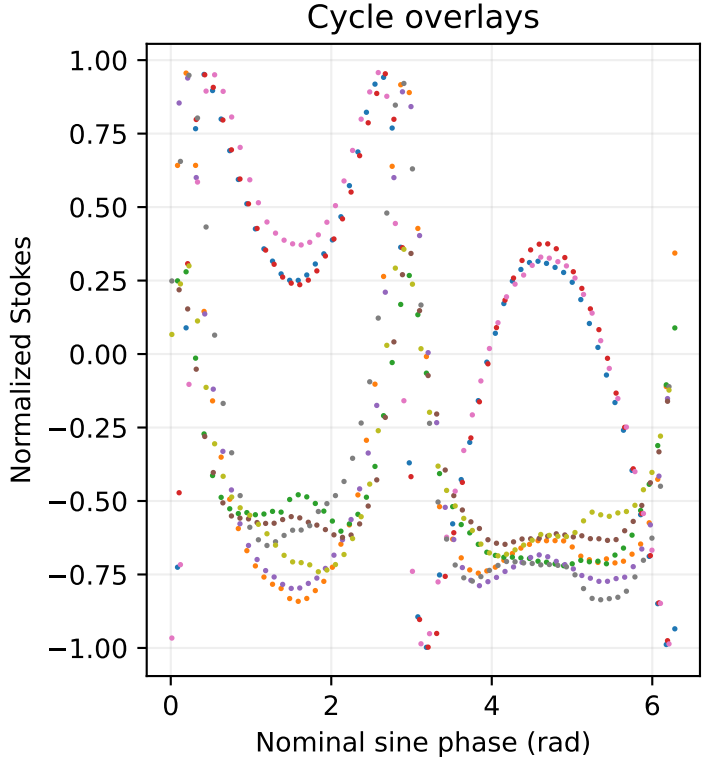
phi2 sine; fixed phi1 = 0.2324 RP V (segment 3)





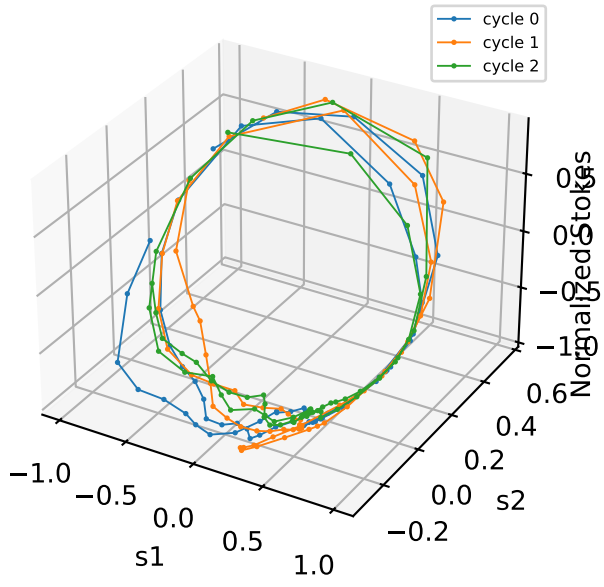


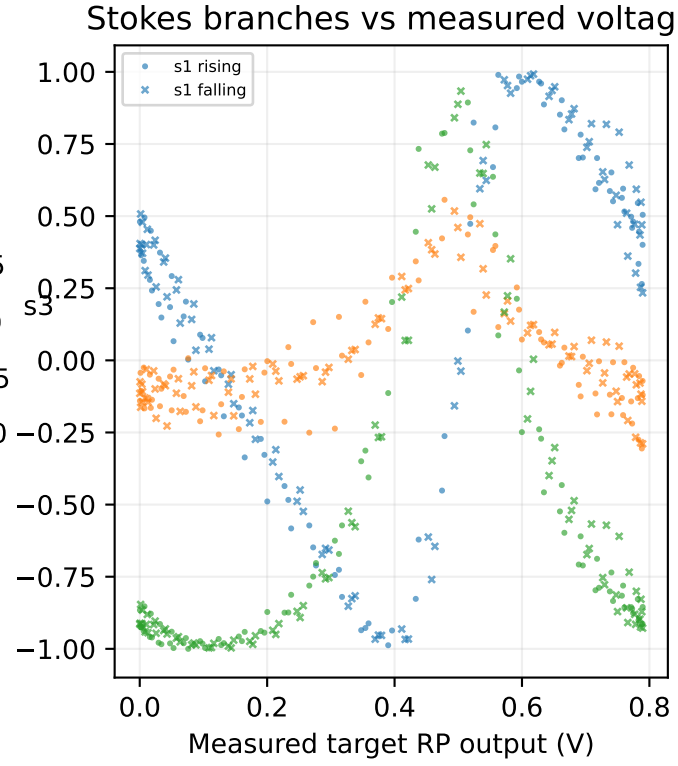


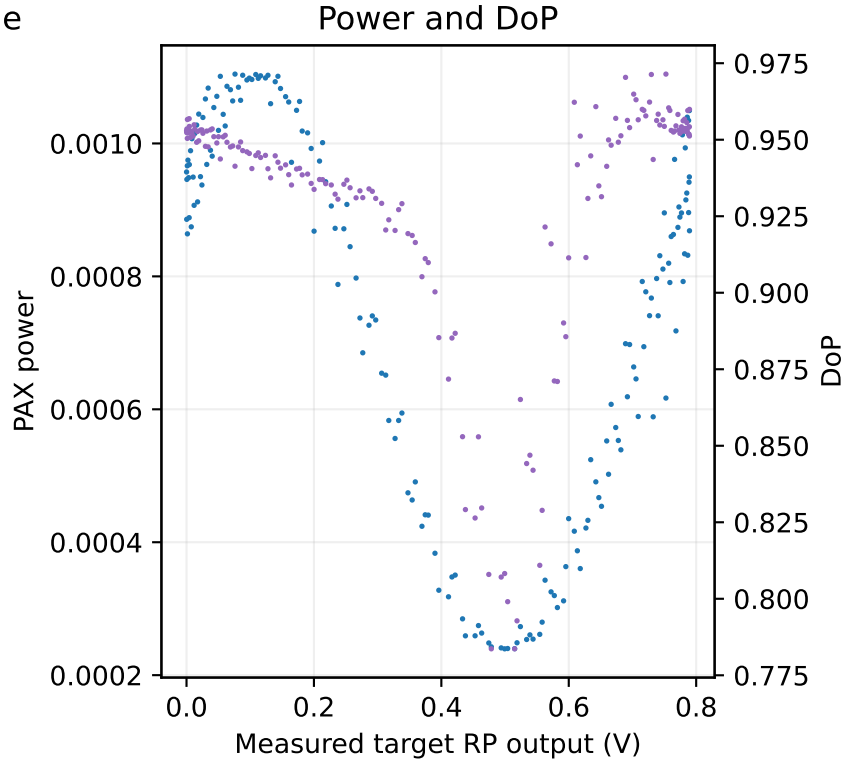


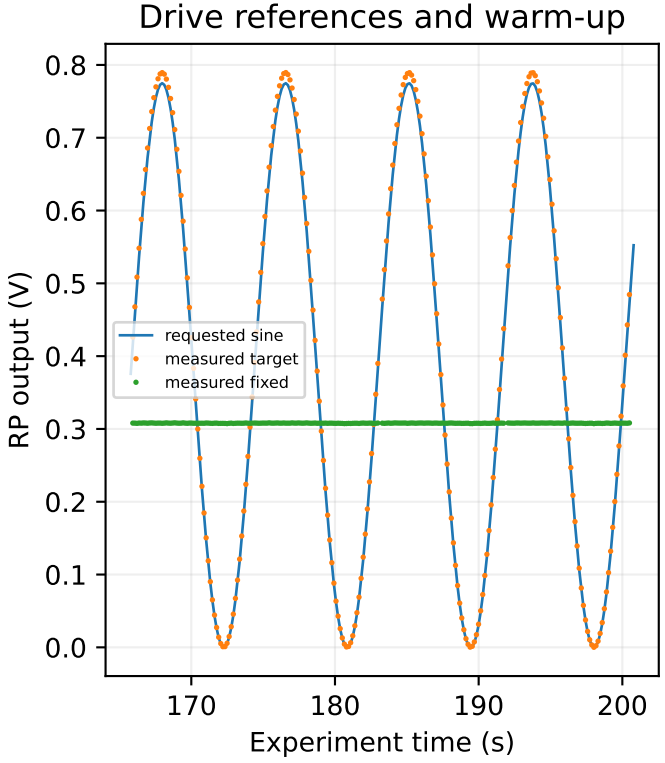
Ideal reference: phi2 changes polar coordinate; fixed phi1 sets azimuth orientation.
Plane normal: [0.121, -0.465, 0.877]
Plane residual RMS: 0.065
Rising/falling angular RMS: 11.2°
Rising/falling power RMS: 3.97e-05 PAX units
Cycle angular RMS: 10°
Cycle power RMS: 4.39e-05 PAX units
Cycle DoP RMS: 0.0115
Tracking RMS: 0.009052 V
Reference coverage: 94.9%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.10, -0.47, 0.88], residual 0.0794
Cycle 1 plane: [0.11, -0.45, 0.89], residual 0.0353
Cycle 2 plane: [0.15, -0.48, 0.86], residual 0.0646

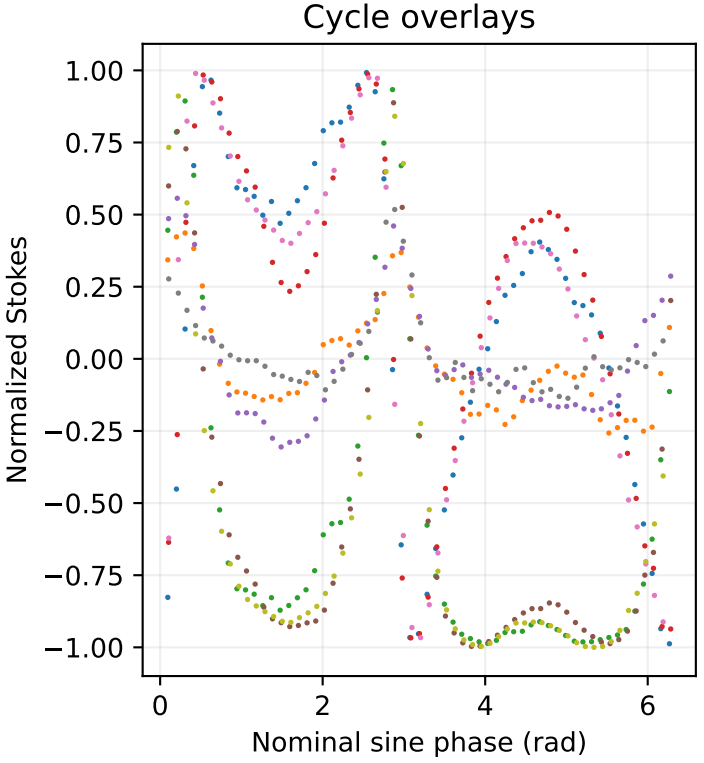
phi2 sine; fixed phi1 = 0.3099 RP V (segment 4)





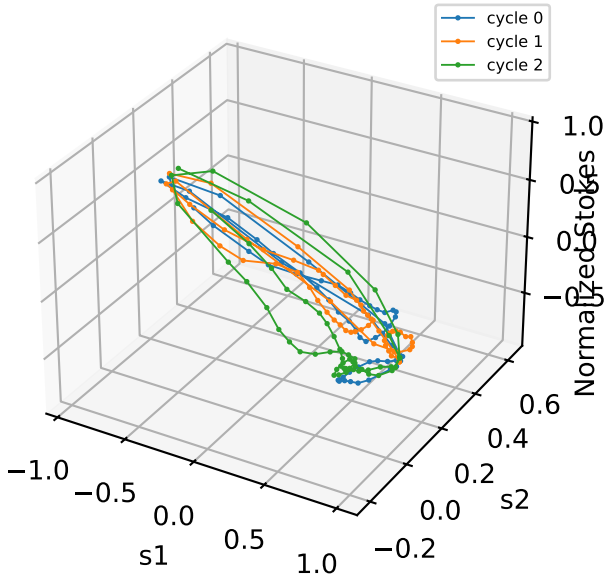


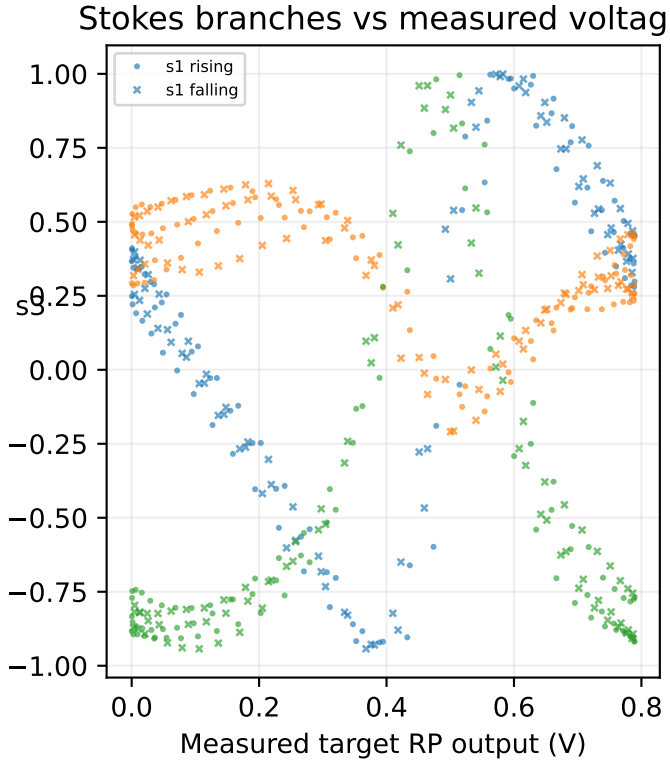


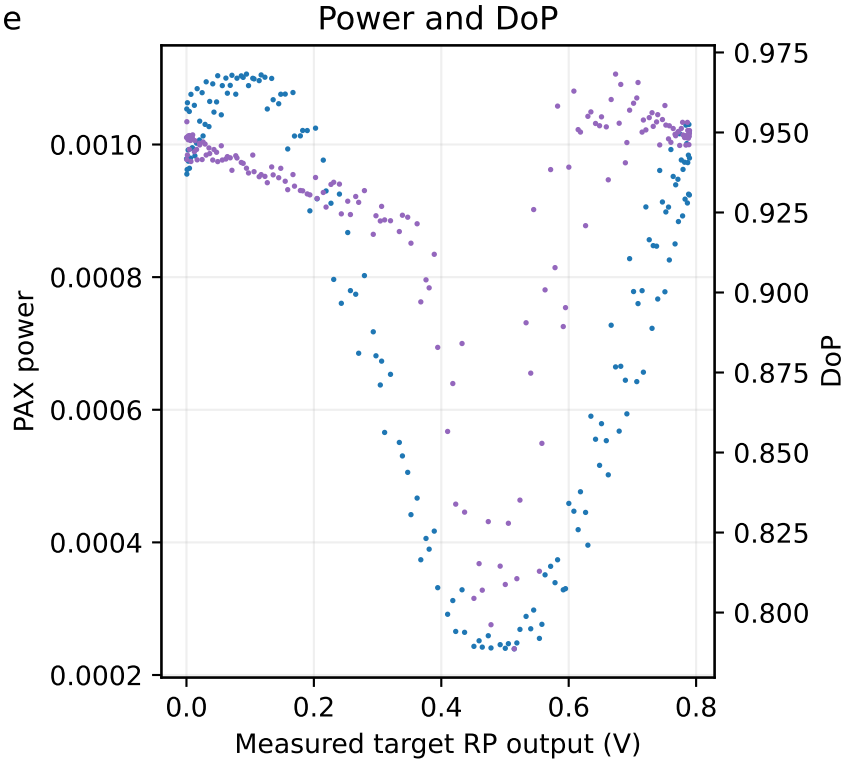


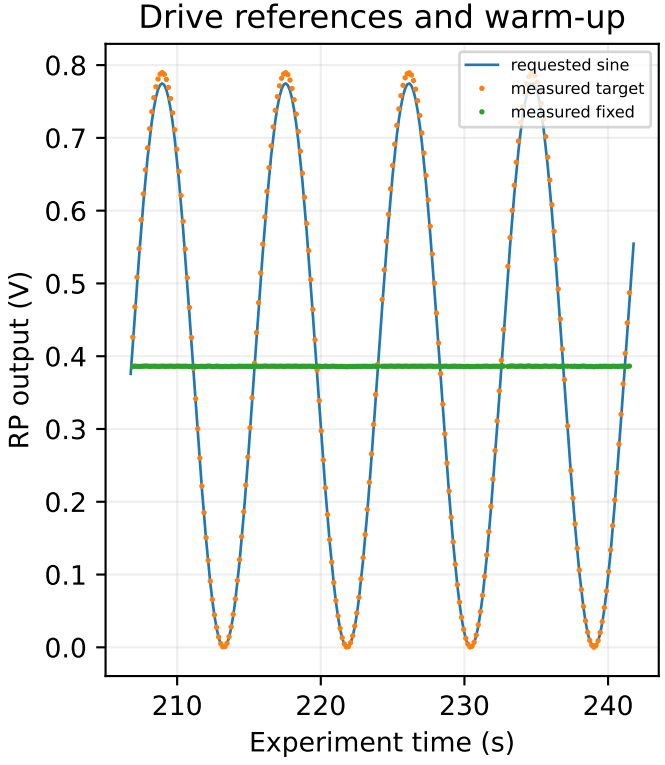
Ideal reference: phi2 changes polar coordinate; fixed phi1 sets azimuth orientation.
Plane normal: [0.031, 0.956, -0.292]
Plane residual RMS: 0.07425
Rising/falling angular RMS: 11.4°
Rising/falling power RMS: 5.79e-05 PAX units
Cycle angular RMS: 13°
Cycle power RMS: 7.91e-05 PAX units
Cycle DoP RMS: 0.0146
Tracking RMS: 0.009211 V
Reference coverage: 98.9%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [-0.01, 0.96, -0.29], residual 0.0668
Cycle 1 plane: [0.08, 0.94, -0.34], residual 0.0693
Cycle 2 plane: [0.02, 0.97, -0.24], residual 0.0513

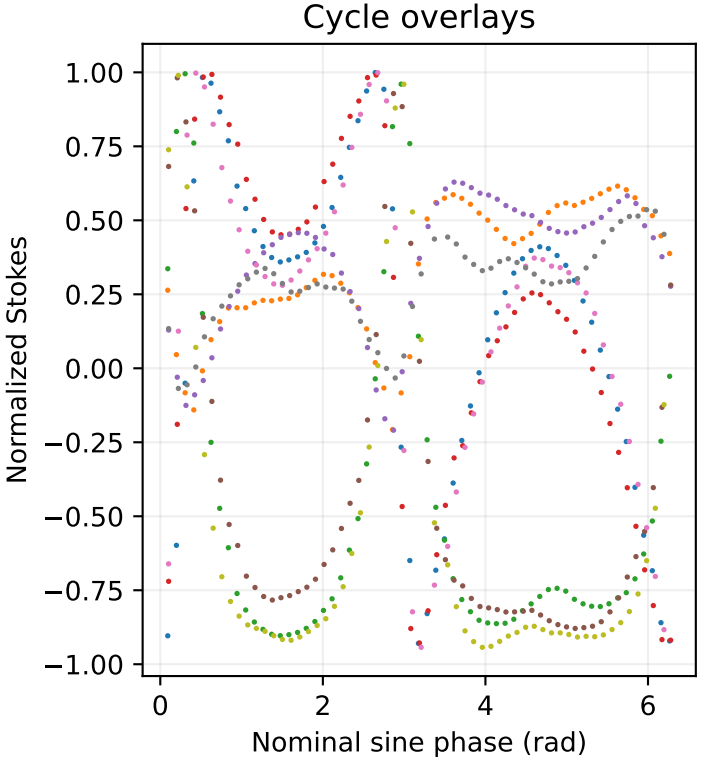
phi2 sine; fixed phi1 = 0.3875 RP V (segment 5)





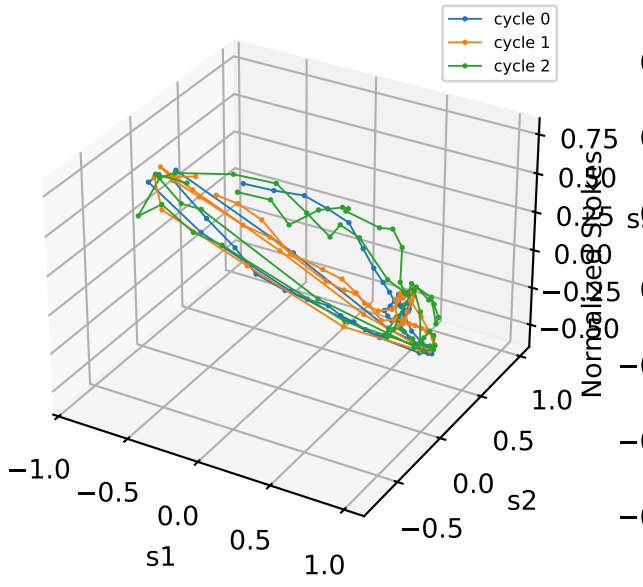


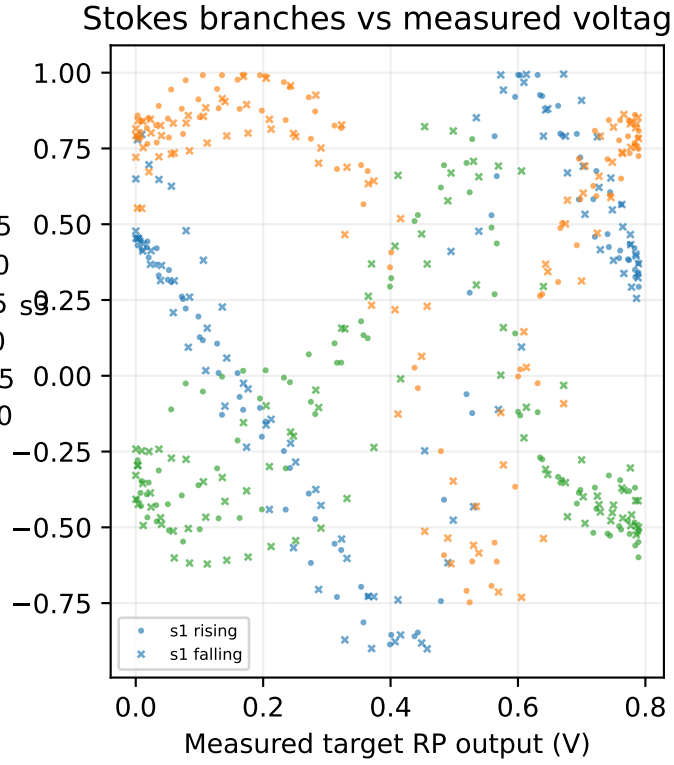


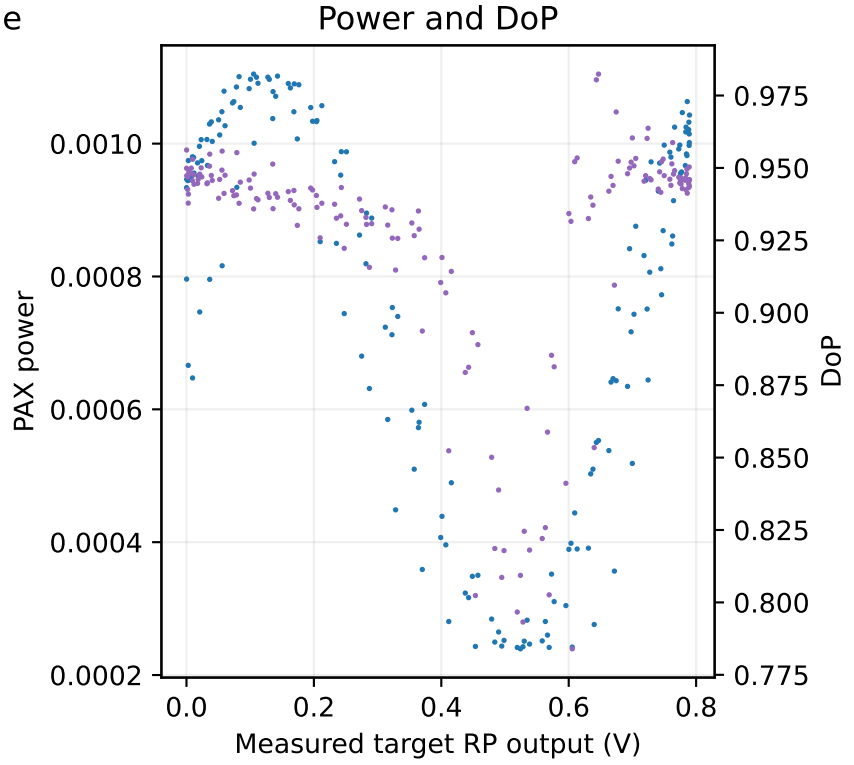


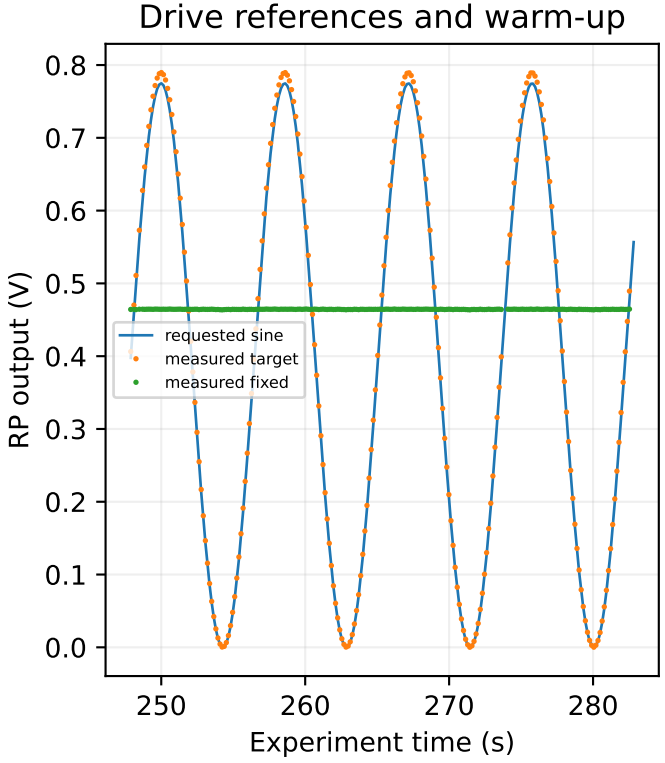
Ideal reference: phi2 changes polar coordinate; fixed phi1 sets azimuth orientation.
Plane normal: [0.197, 0.939, 0.283]
Plane residual RMS: 0.0838
Rising/falling angular RMS: 12.6°
Rising/falling power RMS: 5.28e-05 PAX units
Cycle angular RMS: 12.9°
Cycle power RMS: 5.84e-05 PAX units
Cycle DoP RMS: 0.0145
Tracking RMS: 0.00924 V
Reference coverage: 98.9%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.22, 0.92, 0.32], residual 0.0879
Cycle 1 plane: [0.17, 0.92, 0.35], residual 0.0437
Cycle 2 plane: [0.18, 0.96, 0.20], residual 0.043

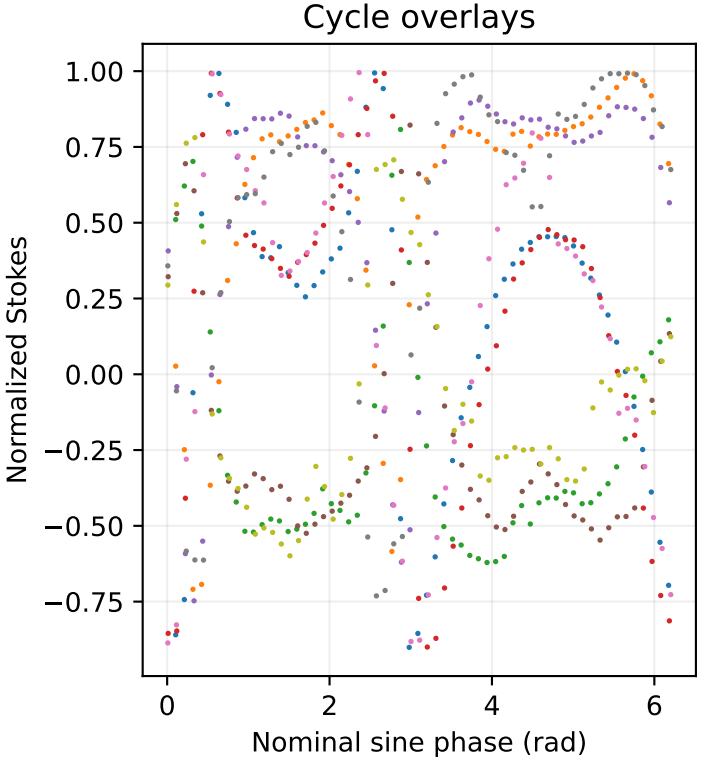
phi2 sine; fixed phi1 = 0.465 RP V (segment 6)





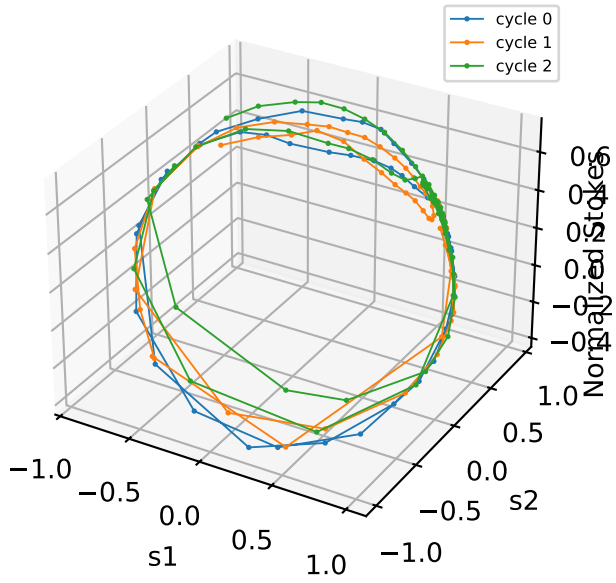


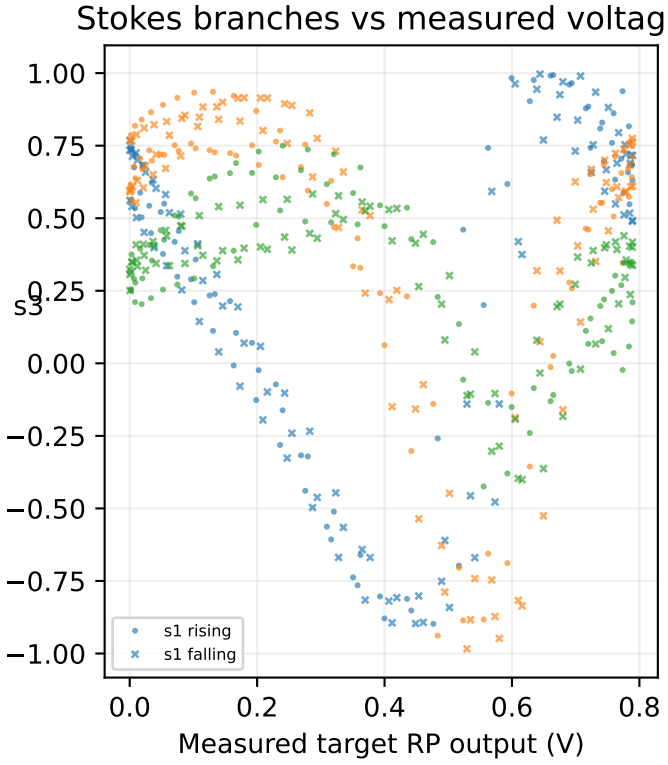


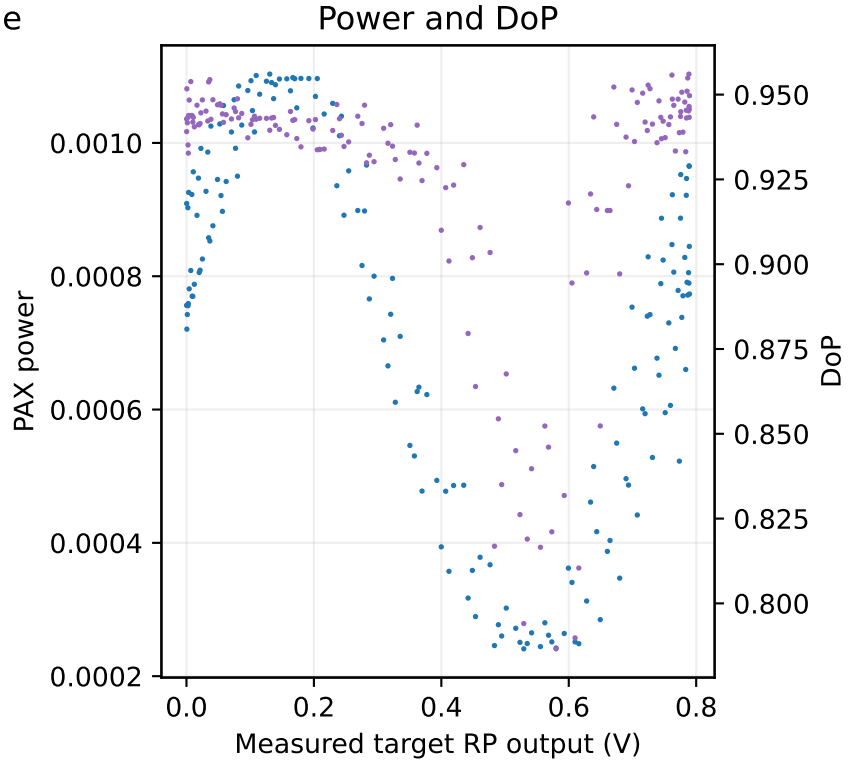


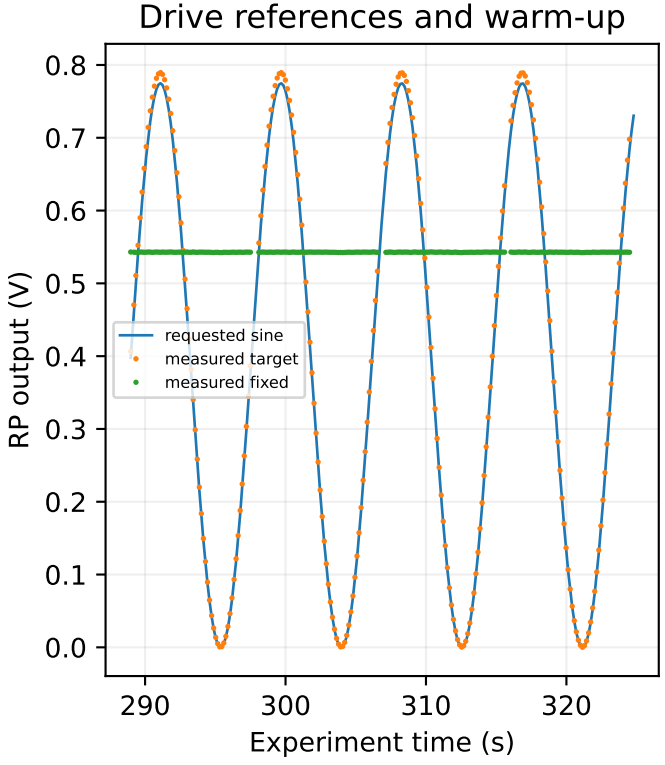
Ideal reference: phi2 changes polar coordinate; fixed phi1 sets azimuth orientation.
Plane normal: [0.284, 0.538, 0.794]
Plane residual RMS: 0.1244
Rising/falling angular RMS: 24.1°
Rising/falling power RMS: 9.3e-05 PAX units
Cycle angular RMS: 23.2°
Cycle power RMS: 0.000116 PAX units
Cycle DoP RMS: 0.0249
Tracking RMS: 0.009231 V
Reference coverage: 98.9%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.27, 0.58, 0.77], residual 0.13
Cycle 1 plane: [0.24, 0.57, 0.78], residual 0.0675
Cycle 2 plane: [0.34, 0.44, 0.83], residual 0.116

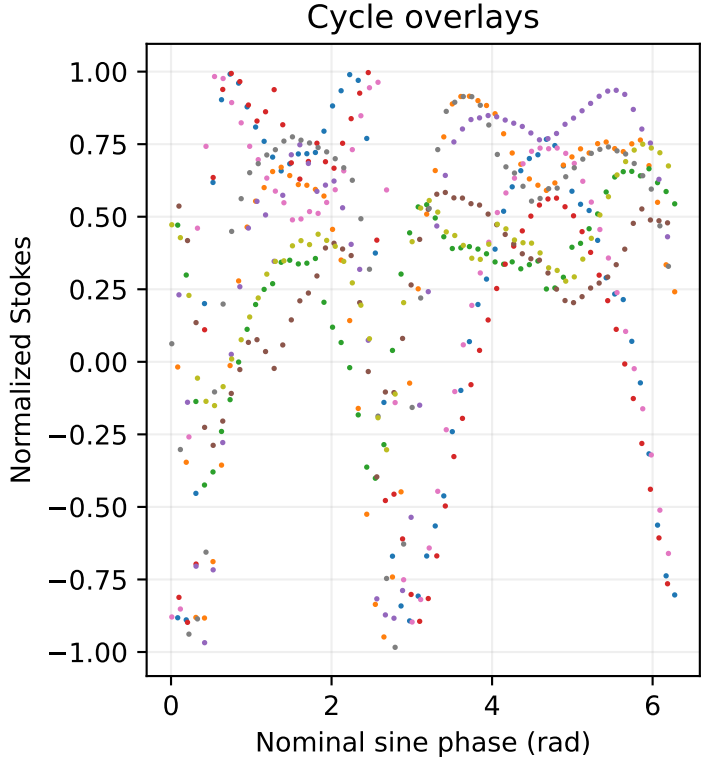
phi2 sine; fixed phi1 = 0.5425 RP V (segment 7)





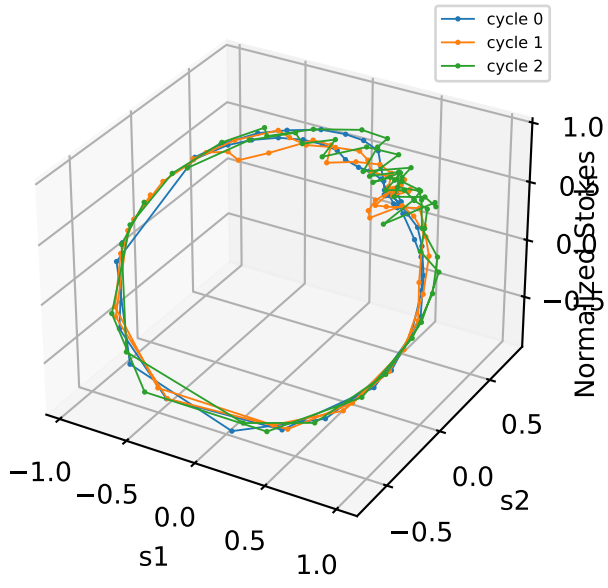


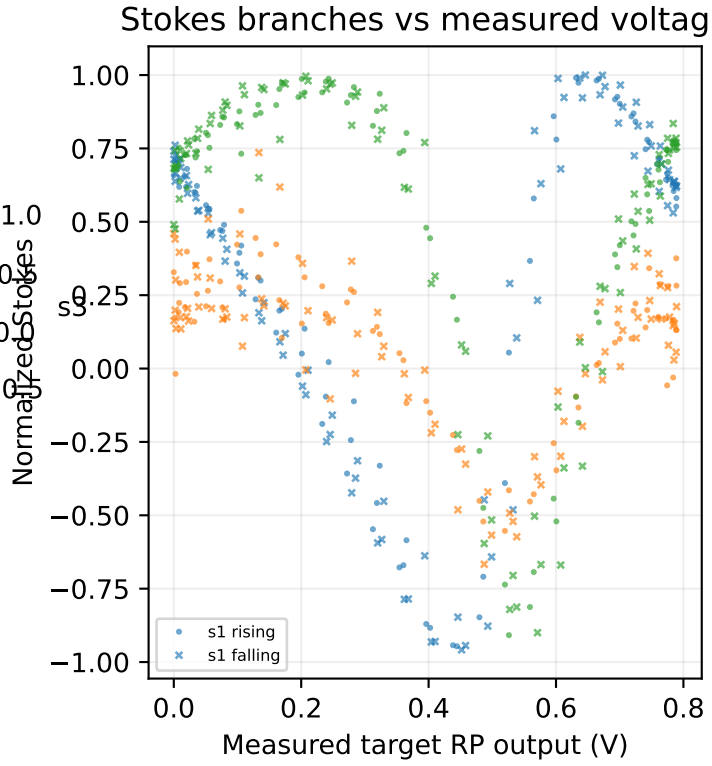


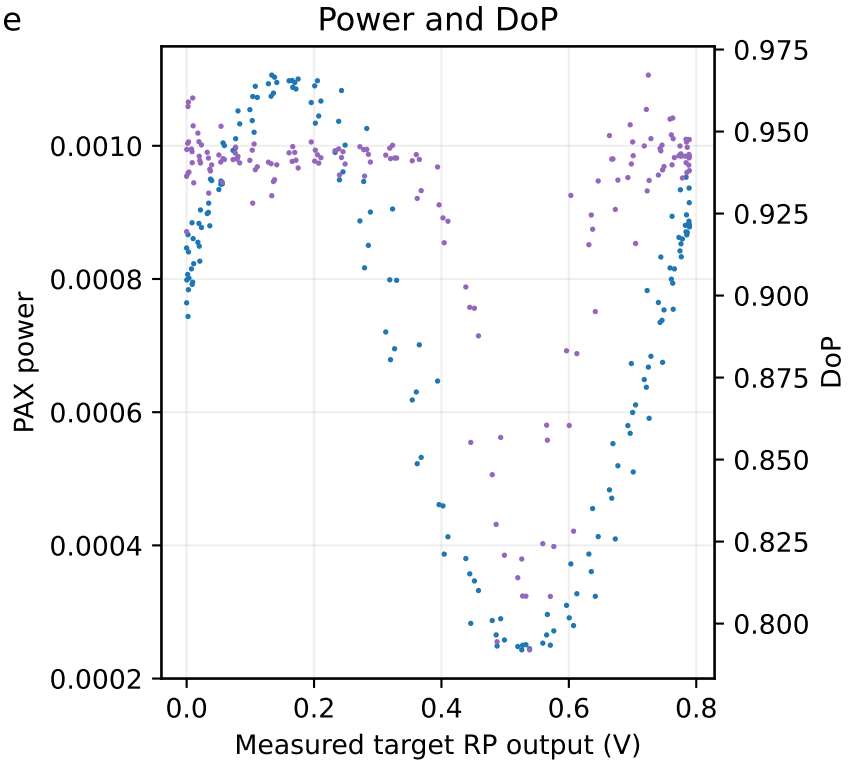


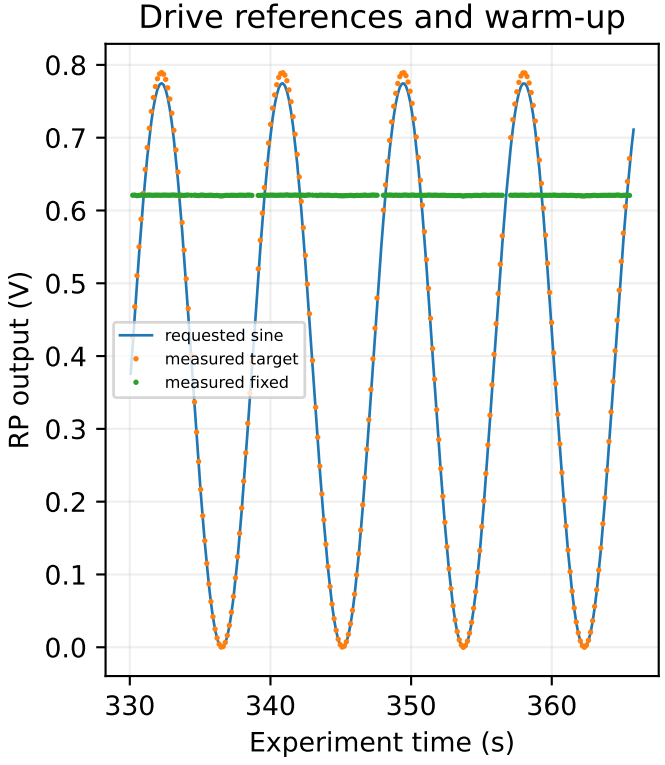
Ideal reference: phi2 changes polar coordinate; fixed phi1 sets azimuth orientation.
Plane normal: [0.255, -0.371, 0.893]
Plane residual RMS: 0.1027
Rising/falling angular RMS: 21.4°
Rising/falling power RMS: 6.71e-05 PAX units
Cycle angular RMS: 20.7°
Cycle power RMS: 0.000116 PAX units
Cycle DoP RMS: 0.0218
Tracking RMS: 0.009036 V
Reference coverage: 95.5%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.24, -0.42, 0.87], residual 0.0853
Cycle 1 plane: [0.25, -0.33, 0.91], residual 0.0889
Cycle 2 plane: [0.28, -0.34, 0.90], residual 0.103

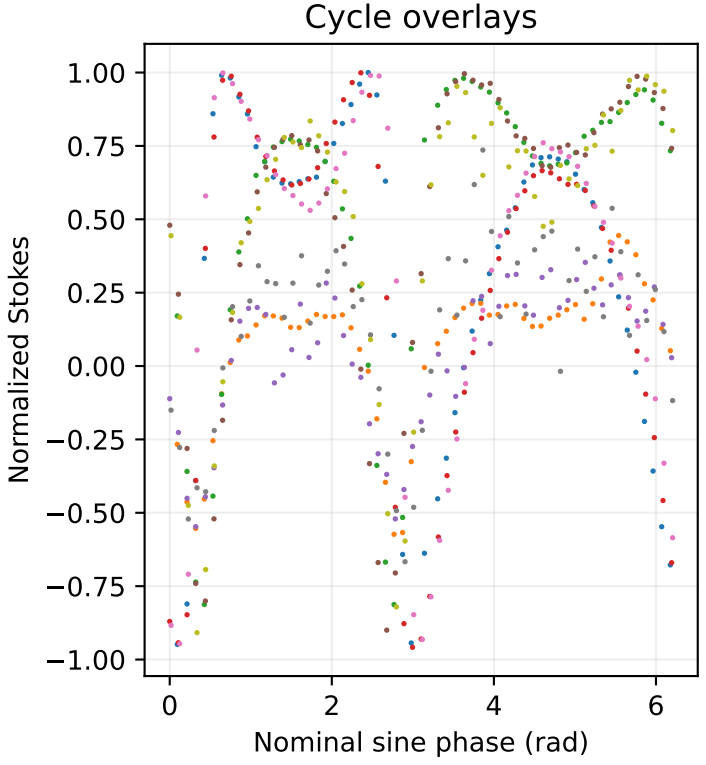
phi2 sine; fixed phi1 = 0.62 RP V (segment 8)





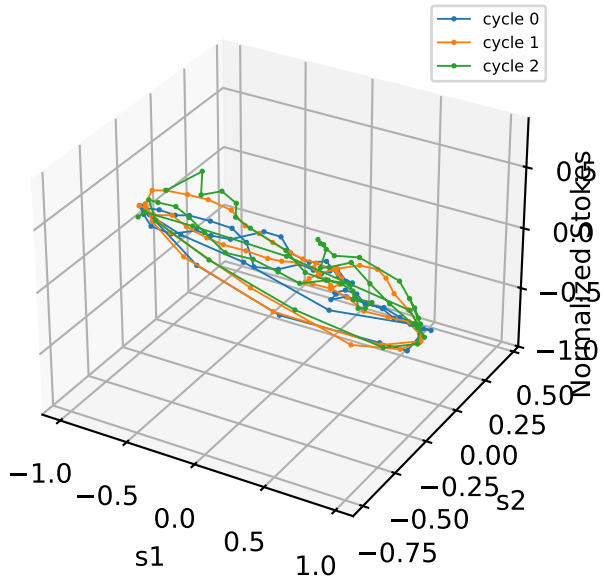


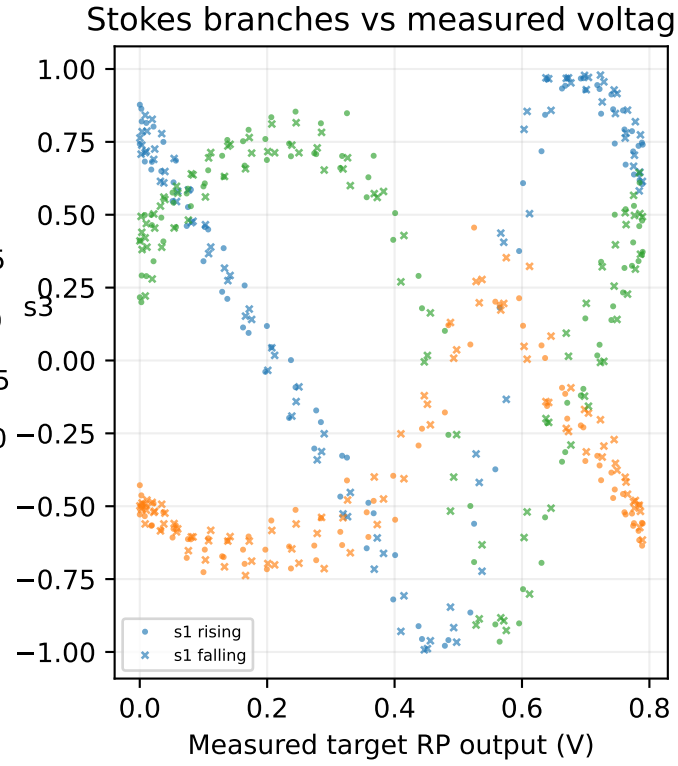


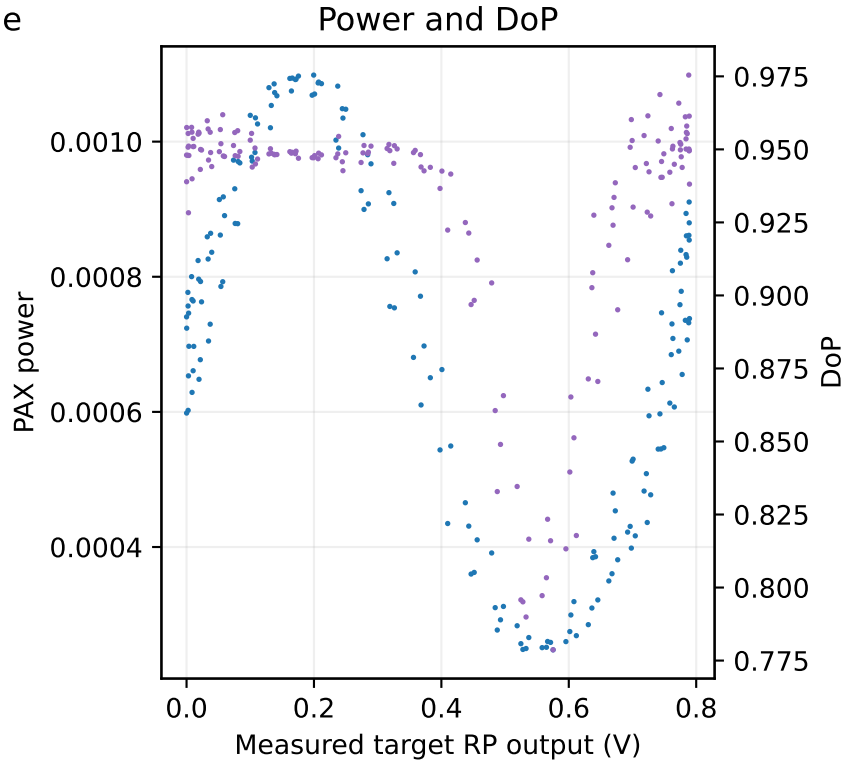


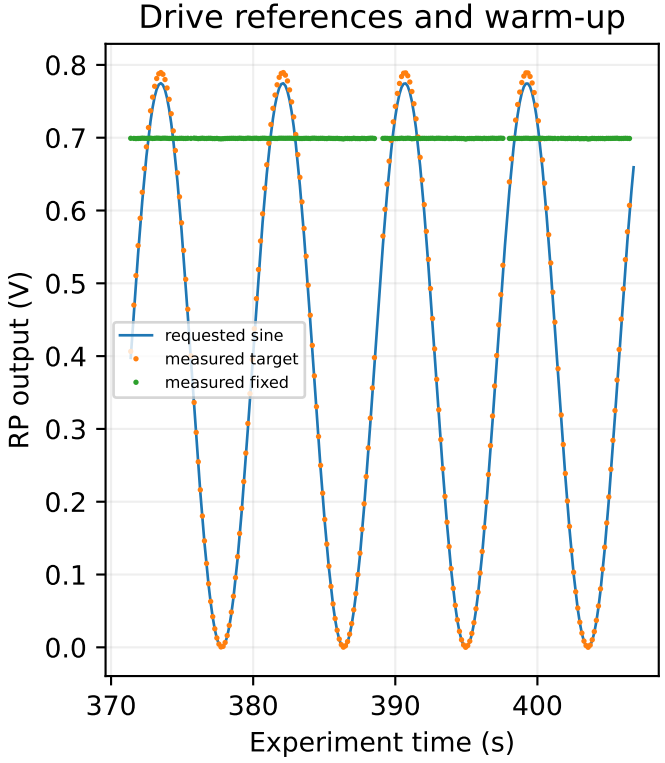
Ideal reference: phi2 changes polar coordinate; fixed phi1 sets azimuth orientation.
Plane normal: [-0.155, 0.903, -0.401]
Plane residual RMS: 0.1129
Rising/falling angular RMS: 13.6°
Rising/falling power RMS: 7.1e-05 PAX units
Cycle angular RMS: 12.7°
Cycle power RMS: 6.44e-05 PAX units
Cycle DoP RMS: 0.0129
Tracking RMS: 0.00902 V
Reference coverage: 96.0%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [-0.11, 0.91, -0.40], residual 0.0643
Cycle 1 plane: [-0.15, 0.92, -0.35], residual 0.0893
Cycle 2 plane: [-0.19, 0.87, -0.45], residual 0.133

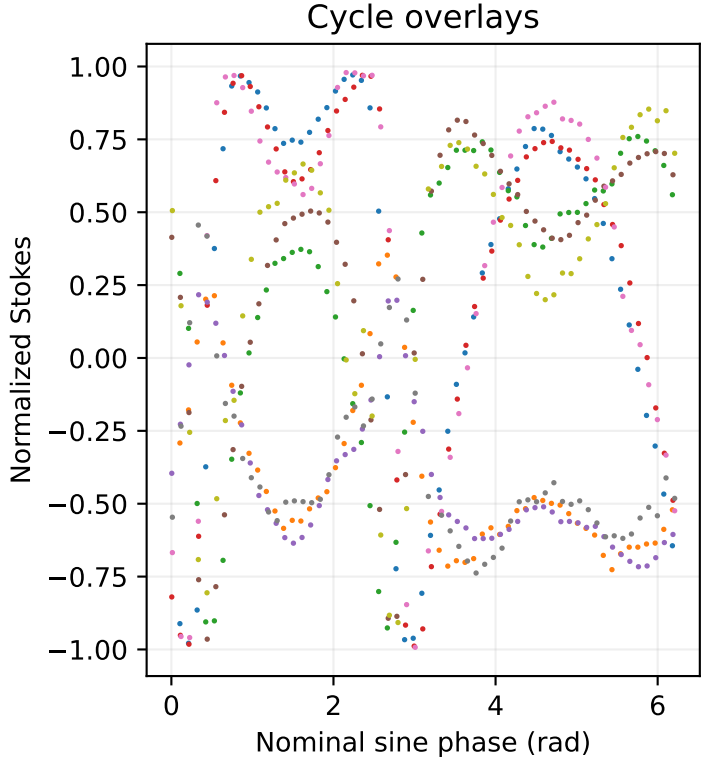
phi2 sine; fixed phi1 = 0.6974 RP V (segment 9)





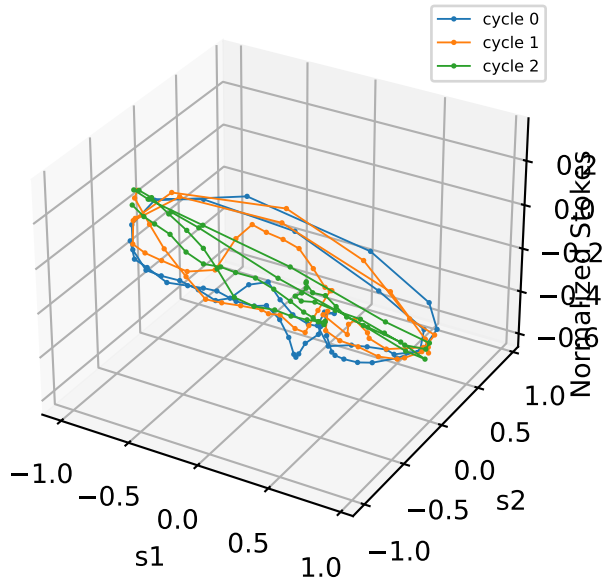




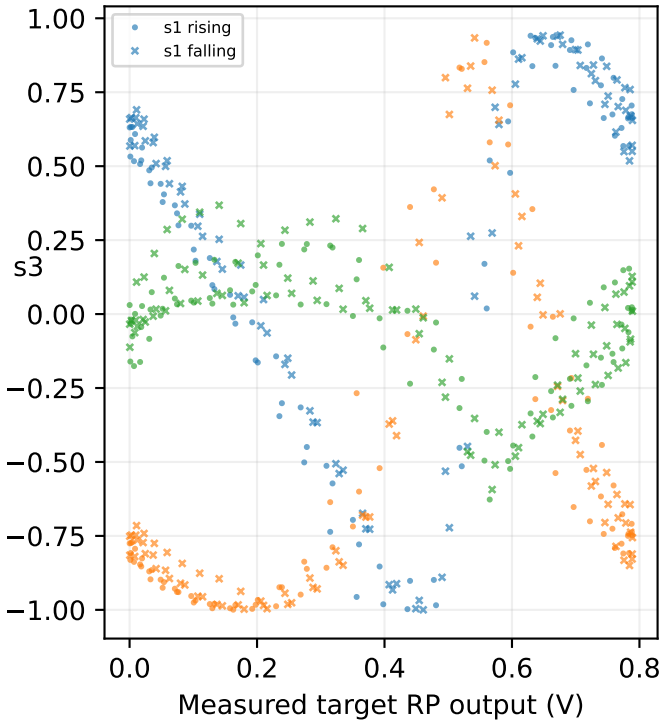


Ideal reference: phi2 changes polar coordinate; fixed phi1 sets azimuth orientation.
Plane normal: [0.067, 0.867, 0.494]
Plane residual RMS: 0.06667
Rising/falling angular RMS: 11.9°
Rising/falling power RMS: 6.47e-05 PAX units
Cycle angular RMS: 15.7°
Cycle power RMS: 8.5e-05 PAX units
Cycle DoP RMS: 0.0179
Tracking RMS: 0.009049 V
Reference coverage: 97.1%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.06, 0.86, 0.51], residual 0.0477
Cycle 1 plane: [0.07, 0.88, 0.47], residual 0.0568
Cycle 2 plane: [0.08, 0.86, 0.50], residual 0.0796

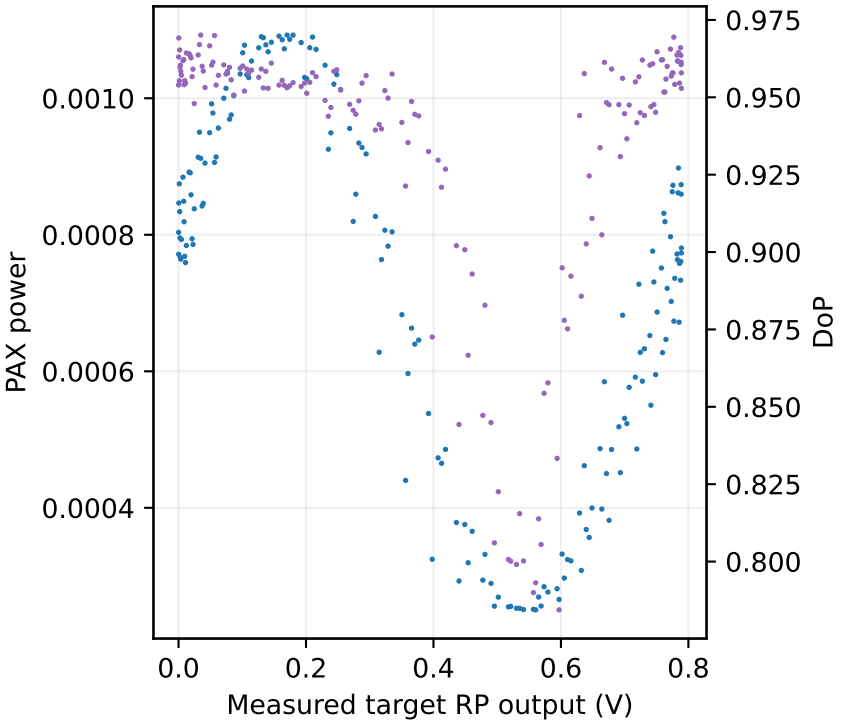
phi2 sine; fixed phi1 = 0.7749 RP V (segment 10)



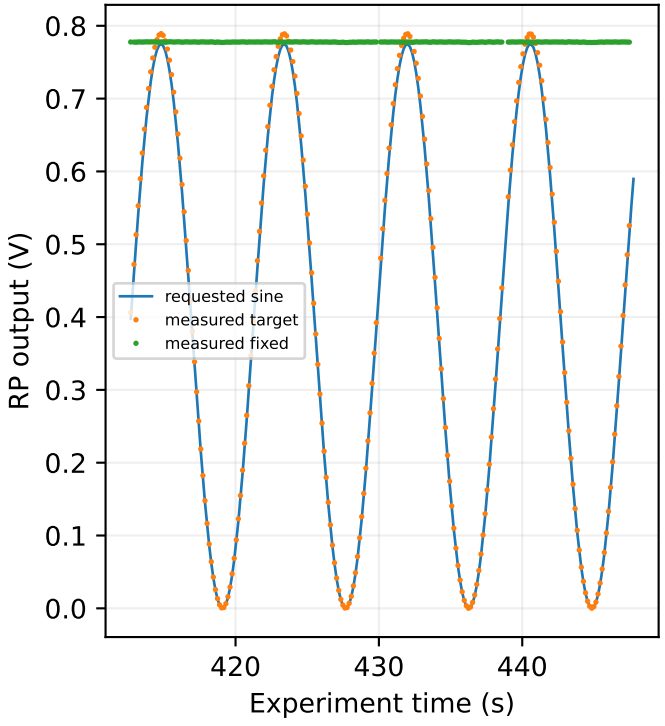
Stokes branches vs measured voltage



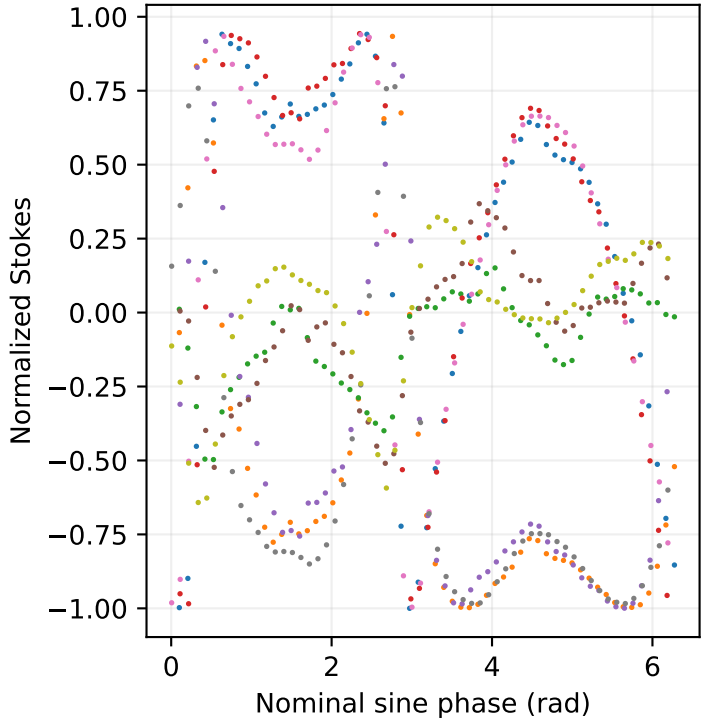
Power and DoP



Drive references and warm-up



Cycle overlays

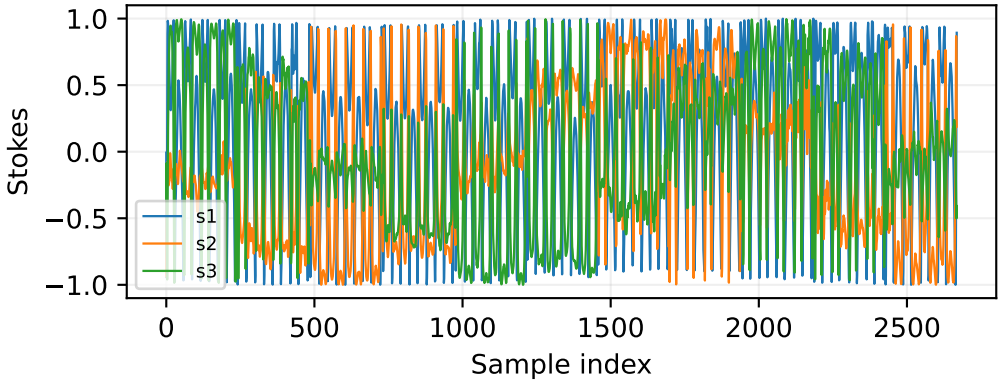


Ideal reference: phi2 changes polar coordinate; fixed phi1 sets azimuth orientation.
Plane normal: [0.142, 0.325, 0.935]
Plane residual RMS: 0.09044
Rising/falling angular RMS: 17.3°
Rising/falling power RMS: 7.7e-05 PAX units
Cycle angular RMS: 12.9°
Cycle power RMS: 6.24e-05 PAX units
Cycle DoP RMS: 0.0141
Tracking RMS: 0.008987 V
Reference coverage: 98.3%
Branch separation is reported descriptively; it is not assigned to piezo hysteresis.
Cycle 0 plane: [0.13, 0.24, 0.96], residual 0.0631
Cycle 1 plane: [0.14, 0.32, 0.94], residual 0.0993
Cycle 2 plane: [0.15, 0.41, 0.90], residual 0.0475

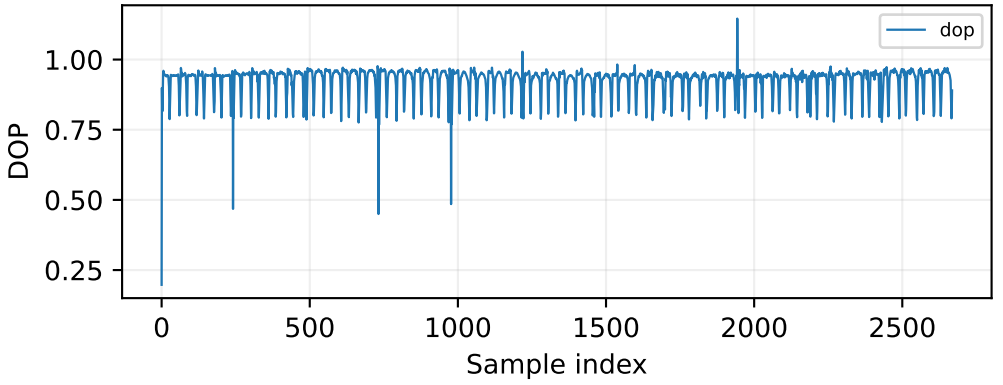
Recorded telemetry

Acquisition order preserves stage changes; no timing alignment or transfer gain is inferred.

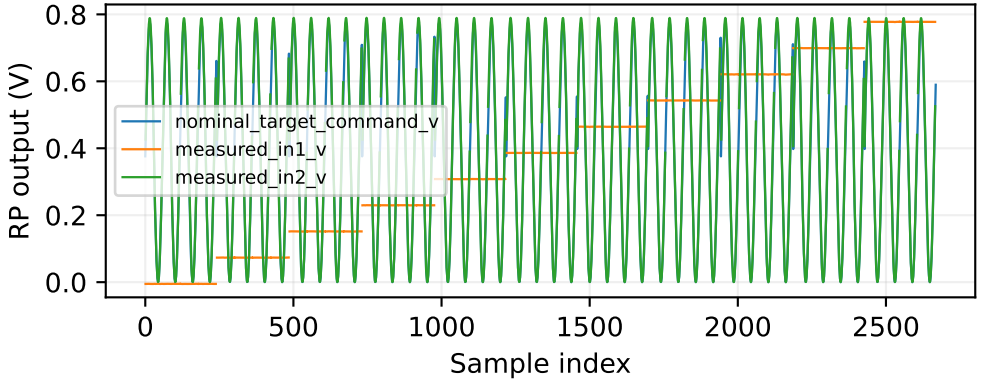
Normalized Stokes



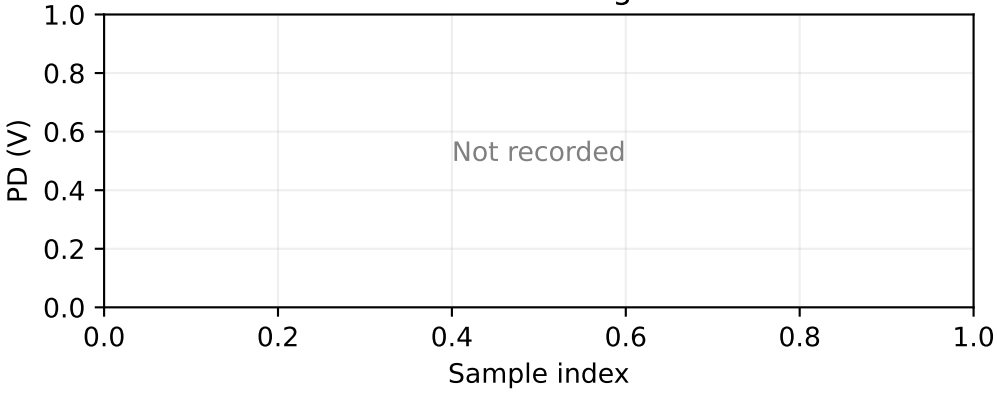
Raw degree of polarization



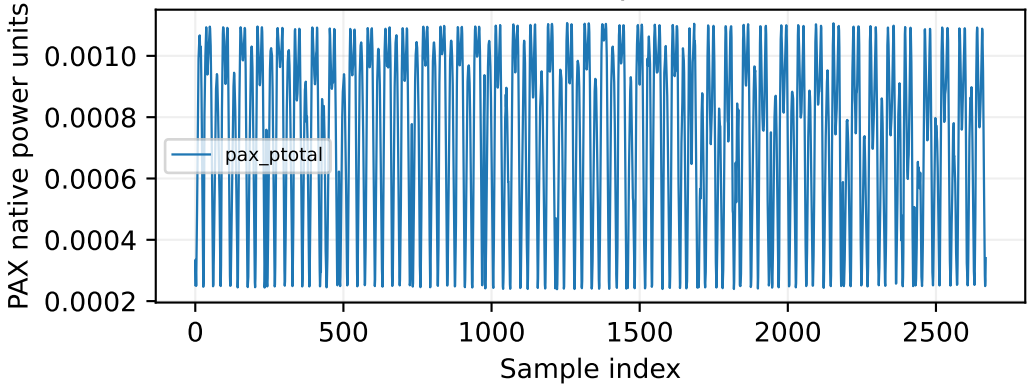
Actuator command / reference history



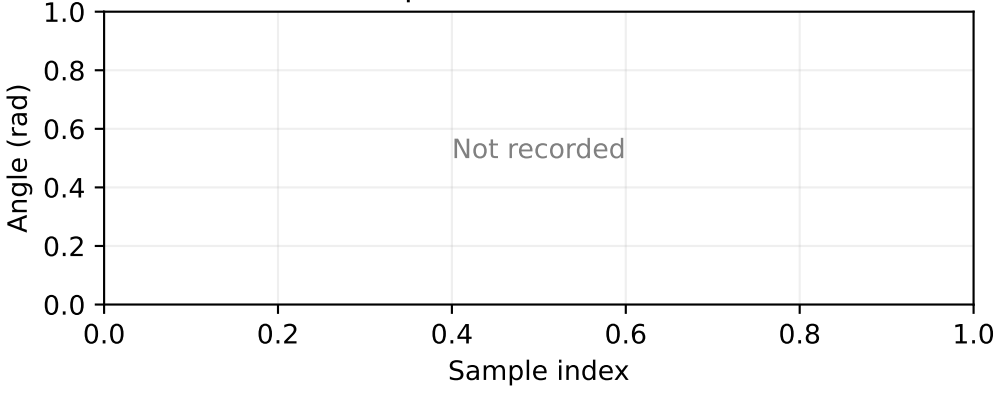
Photodiode signal



PAX total power



Sphere coordinates



Active parameters

RP commands: volts, bounded to 0–1. V_lambda: terminal volts for 2π. Gains: terminal V / RP V.

Parameter	Recorded value
label	"phi2-rerun"
comment	"same as phi1"
actuator	"phi2"
report	"pdf"
pax_rotation_velocity_hz	60.0
pax_measurement_mode	5
pax_wavelength_nm	830.0
phi1_v_lambda	13.078125
phi2_v_lambda	13.078125
phi1_actuator_volts_per_rp_volt	16.875
phi2_actuator_volts_per_rp_volt	16.875
rp_output_min_voltage	0.0
rp_output_max_voltage	1.0
cross_sweep_sine_frequency_hz	0.1
cross_sweep_sine_center_voltage	0.3875
cross_sweep_sine_amplitude_voltage	0.3875
cross_sweep_warmup_cycles	1
cross_sweep_recorded_cycles	3
cross_sweep_phi1_bias_voltages	[0.0, 0.0775, 0.155, 0.2325, 0.31, 0.3875, 0.465, 0.5425, 0.62, 0
cross_sweep_phi2_bias_voltages	[0.0, 0.0775, 0.155, 0.2325, 0.31, 0.3875, 0.465, 0.5425, 0.62, 0
cross_sweep_settle_s	1.0
cross_sweep_bias_ramp_s	1.0
cross_sweep_bias_ramp_updates_per_s	50.0
cross_sweep_reference_sample_rate_hz	400.0

Active parameters

RP commands: volts, bounded to 0–1. V_lambda: terminal volts for 2π. Gains: terminal V / RP V.

Parameter	Recorded value
cross_sweep_scope_block_s	8.0
cross_sweep_min_pax_samples_per_cycle	20.0
cross_sweep_pax_latency_s	0.12